

ADVERSARIAL ATTACKS LEVERAGE INTERFERENCE BETWEEN FEATURES IN SUPERPOSITION

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Anonymous submission

ABSTRACT

Fundamental questions remain about when and why adversarial examples (AExs) arise in neural networks (NNs), with existing accounts often emphasising either irregular decision landscapes or sensitivity to non-robust input features. We reframe this question around a quantitatively testable mechanism: adversarial vulnerability can emerge from *efficient* information encoding in NNs. Specifically, we show that superposition – where networks represent more features than available dimensions – creates measurable arrangements of latent representations that adversaries can exploit. Across controlled settings, adversarial perturbations align with the interference structure induced by superposed features, making attack direction, magnitude, transfer, and class-specific vulnerability predictable from feature geometry. This comparison-based framework explains two empirical patterns already observed in adversarial robustness: attacks transfer most readily between models with similar learned geometry, and vulnerability varies systematically by class. In synthetic experiments with precisely controlled superposition, we establish that superposition *suffices* to create adversarial vulnerability, with projected gradient descent (PGD)-discovered attacks closely matching theoretically optimal perturbations and clearly separating from random baselines. We then show the same qualitative pattern in a vision transformer (ViT) trained on CIFAR-10, where robustness changes consistently with bottleneck-induced superposition. Together, these findings show that adversarial vulnerability can be a quantitatively predictable byproduct of representational compression, rather than only a flaw in learning or a property of non-robust inputs. We release our implementation through a software repository.

1 INTRODUCTION

Despite extensive research on AExs (Szegedy et al., 2014; Goodfellow et al., 2014), their empirical regularities still lack a single mechanistic account. Small input perturbations can produce large prediction changes, yet current explanations often do not specify which perturbations should succeed, why success rates differ across classes, or why attacks transfer more reliably between some pairs of models than others. A useful explanation should therefore connect the qualitative phenomenon of adversarial vulnerability to measurable quantities: feature geometry, perturbation direction, attack success, robust accuracy, and transfer rates.

Existing explanations broadly fit into two camps (Nakkiran, 2019). The *bug* perspective attributes vulnerability to learning flaws (Schmidt et al., 2018; Nakkiran, 2019) or geometric properties of high-dimensional decision boundaries (Fawzi et al., 2016); this line of work often gives bounds on quantities such as capacity (Bubeck & Sellke, 2021) or Lipschitz constants (Hein & Andriushchenko, 2017), but it typically does not identify the successful perturbation pattern for a given example. The *feature* perspective argues that AExs exploit predictive but non-robust statistical patterns in data (Ilyas et al., 2019), shifting attention to the relation between data statistics and human perception. However, it often treats non-robust features as properties of the dataset rather than as quantities shaped by a model’s encoding strategy. As a result, neither view fully explains how representational constraints, data semantics, and observed attack metrics interact.

We bridge this gap through a mechanistic interpretability account in which adversarial vulnerability is evaluated as an architecture–semantics interaction with quantitative consequences. We show that AExs can emerge from interference between learned representations in NNs. In this account, adversarial attacks systematically exploit superposed latent structure – a mechanism that normally

increases representational capacity – to produce perturbations whose signs, magnitudes, and transfer patterns are determined by latent geometry. The resulting pathway is empirically checkable: input correlations constrain feature arrangements, feature arrangements determine interference patterns, and interference patterns predict attack characteristics and transferability. This makes the comparison structure explicit, allowing perturbations discovered by PGD, theoretically optimal perturbations, and random baselines to be evaluated against the same geometric quantities.

Our account draws on recent theories of network representation, especially the linear representation hypothesis (LRH) (Park et al., 2024) and the theory of superposition (Elhage et al., 2022). The LRH posits that *input features*—fundamental abstractions of data—are represented as linear directions in a network’s representation space. Superposition then explains how NNs can represent substantially more features than they have neurons by packing multiple feature directions into shared dimensions, gaining capacity while introducing measurable interference. This paper asks whether that interference is not merely a representational nuisance but a source of adversarial vulnerability, and whether its quantitative structure explains which attacks succeed. Our primary contributions are:

- We present a framework that links architectural constraints, feature properties, and adversarial metrics. In simple models, we show that data properties induce specific superposition geometries, and that those geometries directly determine realised adversarial perturbations, transferability, and class-specific susceptibility.
- We show that the same empirical pattern persists in a ViT trained on CIFAR-10 with an engineered bottleneck layer that induces superposition, foregrounding how robust accuracy changes under controlled representational compression.
- We show that superposition is sufficient, but not necessary, for adversarial vulnerability. This comparison separates superposition-driven interference from a second mechanism, algorithmic brittleness, while preserving a shared representation-based framing.
- We demonstrate that mechanistic knowledge of learned representations enables informed adversarial attack construction, making the practical significance of the measured geometry clear for both attacks and semantically informed defences.

2 BACKGROUND

We use the background concepts in this section to define the quantities compared throughout the paper: latent feature directions, their degree of non-orthogonality, and the perturbation budgets used to measure adversarial robustness. This framing lets us treat attack success not only as a binary outcome, but as a pattern that can be compared against the geometry learned by the model.

Linear representation hypothesis (LRH). The LRH states that NNs often encode variables used in their computation, including semantic properties of inputs, as linear directions in activation space that can be analysed as abstractions (Park et al., 2024; Guerner et al., 2023). This linear framing is useful for quantitative comparison because a feature’s effect can be measured by direction, magnitude, and alignment with downstream weights. Linear separability also makes features easy for subsequent layers to recognise and manipulate through dot products, and growing empirical work supports this picture (Gurnee et al., 2023; Park et al., 2024). Let $\mathcal{C} = \{c_1, \dots, c_M\}$ denote a set of M semantically meaningful latent features (e.g. concepts like “presence of shape,” or “indoor vs. outdoor”). Formally:

Definition 1 (Linear Representation Hypothesis (LRH)) A neural network layer with activations $\mathbf{h}^{(l)} \in \mathbb{R}^{d_l}$ satisfies the LRH if it represents the latent features $\mathcal{C} = \{c_1, \dots, c_M\}$ as linear directions $\{\mathbf{v}_j\}_{j=1}^M \subset \mathbb{R}^{d_l}$ such that:

$$\mathbf{h}^{(l)}(\mathbf{x}) \approx \sum_{j=1}^M a_j(\mathbf{x}) \mathbf{v}_j$$

where $a_j(\mathbf{x}) \in \mathbb{R}$ represents the activation magnitude of feature c_j , and $\mathbf{v}_j$ is the corresponding linear direction. The features are **linearly accessible** (Costa et al., 2025): inputs $\mathbf{x}_0, \mathbf{x}_1 \in \mathcal{X}$ that differ mainly in the value of feature c_j while holding other features approximately constant (i.e., $a_j(\mathbf{x}_1) - a_j(\mathbf{x}_0) = k$ and $|a_i(\mathbf{x}_1) - a_i(\mathbf{x}_0)| < \lambda$ for all $i \neq j$ and some small $\lambda > 0$) satisfy:

$$\mathbf{h}^{(l)}(\mathbf{x}_1) - \mathbf{h}^{(l)}(\mathbf{x}_0) \approx k \mathbf{v}_j$$

where k reflects the change in c_j .

Superposition and sparsity. NNs can represent many more latent features than available activation dimensions, so that $M > d_l$. For example, LLMs can reference many more place names than they have residual stream dimensions. Superposition provides the quantitative lens for this compression: networks use an overcomplete, non-orthogonal set of feature directions $\{\mathbf{v}_j\}_{j=1}^M$ whose pairwise alignments determine interference. The feasibility of this packing follows from the fact that $2^{\Theta(d\epsilon^2)}$ almost orthogonal vectors ($< \epsilon$ cosine similarity) can be represented in d -dimensional space (Tkocz, 2012), and that sparse vectors can be accurately recovered after projection into lower-dimensional spaces (Elhage et al., 2022; Bereska & Gavves, 2024; Sawmya et al., 2025). This efficiency introduces two measurable costs. First, *polysemanticity* (Scherlis et al., 2022; Lecomte et al., 2023) appears when individual neurons contribute to multiple features, weakening one-to-one neuron interpretations. Second, non-orthogonal feature directions create *interference* between features: activating one feature also changes the activation readout of others. Networks can partially manage these costs through non-linear operations (e.g. ReLU, softmax) that disambiguate superposed features (Gurnee et al., 2023), but the residual interference remains the central quantity we test against adversarial perturbations.

Definition 2 (Superposition Hypothesis) *A network layer represents features in superposition if:*

1. *The number of latent features exceeds layer dimensionality: $M > d_l$*
2. *There exists non-orthogonal feature directions: $\exists i, j$ with $i \neq j$ such that $\mathbf{v}_i \cdot \mathbf{v}_j \neq 0$*
3. *Latent feature activations are sparse: $\mathbb{E}_{\mathbf{x}}[\|\mathbf{a}(\mathbf{x})\|_0] \ll M$, where $\mathbf{a}(\mathbf{x}) = [a_1(\mathbf{x}), \dots, a_M(\mathbf{x})]^T$*

The network trades representational capacity against feature interference by packing more features than dimensions.

Adversarial attacks. We use the standard perturbation-budget definition of adversarial attacks and robustness (Szegedy et al., 2014; Costa et al., 2024), which lets us compare clean predictions, attacked predictions, and robust accuracy under the same ℓ_p constraints:

Definition 3 (Adversarial attack) *Let $\mathbb{B}_p(\mathbf{x}, \epsilon)$ denote the ϵ -ball around $\mathbf{x} \in \mathcal{X}$ in the ℓ_p -sense, e.g. $\mathbb{B}_\infty$ is the ℓ_∞ -ball. Any small δ that p -perturbs $\mathbf{x}$ to create an **adversarial input** $\mathbf{x}' \in \mathbb{B}_p(\mathbf{x}, \epsilon)$ such that $f(\mathbf{x}') \neq f(\mathbf{x})$ (untargeted) or $f(\mathbf{x}') = y_{\text{target}}$ (targeted) is known to be an **adversarial attack**. ℓ_∞ - or ℓ_2 -norm attacks are created via the PGD (detailed in Sec. A).*

Definition 4 (Adversarial robustness) *A classifier $g_w(\cdot)$ with parameters w is said to be **robust** at $\mathbf{x}$ with radius $\epsilon > 0$, if $g_w(\mathbf{x}') = g_w(\mathbf{x}) \forall \mathbf{x}' \in \mathbb{B}(\mathbf{x}, \epsilon)$.*

3 SUPERPOSITION GEOMETRY CAN DETERMINE ADVERSARIAL ATTACKS

We begin with the setting where the quantitative predictions are sharpest: a controlled model in which the amount of superposition, the latent feature geometry, and the attack budget can all be measured directly. The experiments compare perturbation profiles, geometric alignments, robust accuracy, and transfer rates while asking three questions:

1. *Do adversarial perturbations exploit the interference between superposed features?*
2. *Do correlations in the input shape the geometric arrangement of superposed latent features?*
3. *Can shared latent geometry explain why attacks transfer between independently trained models?*

Experimental setup. To answer these questions, we use a synthetic task that makes the relevant comparisons explicit while instantiating established LRH and superposition phenomena (Park et al., 2024; Elhage et al., 2022). The task is designed to: (1) give an intuitive classification benchmark for measuring AExs; (2) represent class concepts as linear directions, as assumed by the LRH; (3) induce a controlled gap between the number of class features and the bottleneck dimension; and (4) retain ground-truth knowledge of how each input coordinate maps to a superposed feature, enabling direct quantitative tests of the proposed adversarial mechanism.

Concretely, we partition each input $\mathbf{x} \in \mathbb{R}^d$ into k equally sized groups $\mathbf{x} = [\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)}]$, where $\mathbf{x}^{(j)}$ contains the p input features associated with class j . The label is the group with the largest sum, $y = \operatorname{argmax}_{j \in \{1, \dots, k\}} \sum_{i=1}^p x_i^{(j)}$, so every coordinate has a known class-level contribution. Intuitively, each $x_i^{(j)}$ is evidence for class j , and the p features capture different class attributes.

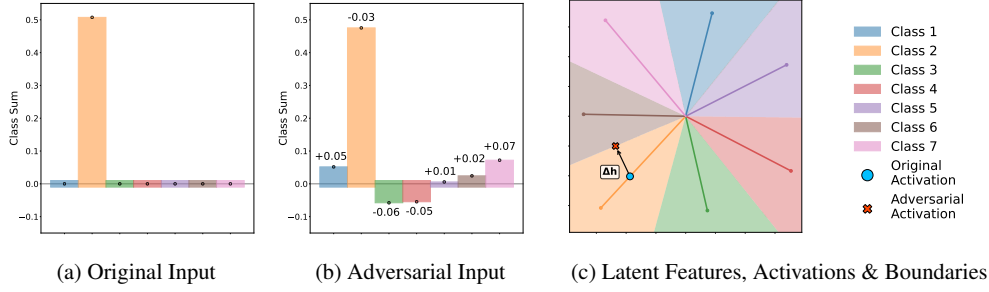


Figure 1: **A representative adversarial attack whose perturbation profile is explained by superposition geometry ($k = 7, m = 2$).** (a) The original sample and its class evidence. (b) The adversarially perturbed sample preserves the ground-truth class but changes the model prediction; the displayed signs and magnitudes show that the perturbation is not a simple increase of the target-class inputs. (c) In activation space, the same perturbation moves the sample across the decision boundary in the direction predicted by the latent feature vectors, shown as the columns of $\mathbf{W}_e$.

Following Elhage et al. (2022), we sample $x_i^{(j)} \sim \text{Uniform}(0, 1)$ with sparsity S (probability of setting to zero), which lets us vary the representational pressure while preserving the same evaluation rule.

A two-layer network compresses the input into m dimensions through a bottleneck encoder, $\mathbf{h} = \text{ReLU}(\mathbf{W}_e \mathbf{x} + \mathbf{b}_e) \in \mathbb{R}^m$ (where $m < k < d$), and reconstructs class logits with a decoder, $\mathbf{z} = \mathbf{W}_d \mathbf{h} + \mathbf{b}_d \in \mathbb{R}^k$. Our primary setup uses cross entropy (CE) loss without ReLUs or biases, chosen because the softmax gives a simple setting in which interference is limited by exponentially suppressing contributions from competing latent features. To check that the empirical pattern is not an artefact of this loss or architecture choice, we repeat the analysis with a second setup using mean squared error (MSE) loss with ReLUs and biases. In both cases, the core quantitative comparison is the same: the model must compress $d = k \times p$ input dimensions into $m < k$ latent dimensions and retain enough class evidence to recover the largest-sum group.

Unlike non-synthetic tasks, where latent features must first be discovered and interpreted, this setup gives ground-truth class labels for the latent features. We can therefore recompute the true class after every perturbation by recalculating group sums, so an AEx is counted only when both evaluation conditions hold:

1. The model’s prediction under perturbation changes (*e.g.* from class A to B)
2. The true class remains unchanged (*e.g.* still class A)

This evaluation separates genuine AExs from perturbations that merely change the ground-truth label (*i.e.* change which group has the highest sum). We generate attacks with ℓ_∞ - and ℓ_2 -norm PGD, keeping the norm constraint explicit for comparisons of robust accuracy and perturbation structure.

Evaluation metrics. We treat the columns of the encoder weight matrix $\mathbf{W}_e$ as the latent representations $\mathbf{v}_i$ for each input feature x_i . Because input feature $x_i^{(j)}$ contributes only to class j , the corresponding $\mathbf{v}_i$ (for $i \in \text{class } j$) provide a class-aligned basis for comparing input perturbations to latent-space movement. This direct mapping lets us measure how perturbations to $\mathbf{x}^{(j)}$ change the representation and whether those changes match the predicted interference pattern. For successful PGD-generated AExs $\mathbf{x}_{adv} = \mathbf{x}_{orig} + \delta$, we report:

1. **Input perturbation profile (IPP):** The sign and magnitude of each δ_i , describing how different input features are perturbed in an attack.
2. **Latent attack alignment:** The similarity $\Delta \mathbf{h} \cdot \mathbf{v}_c$ between the *latent attack vector* $\Delta \mathbf{h} = \mathbf{h}_{adv} - \mathbf{h}_{orig}$ and class c ’s representation.
3. **Attack transferability:** Success rate of attacks generated on one model when applied to another.
4. **Robust accuracy:** The fraction of examples that remain correctly classified after perturbation δ with $\|\delta\|_p \leq \epsilon$ to a sample of inputs.

3.1 RESULTS

Do attacks exploit interference between latent features in superposition? An attack changes the prediction by moving a sample across a decision boundary, so the central quantitative question is

which input coordinates receive positive or negative budget and how large those changes are. Two baselines frame the comparison: the *feature intuition* predicts that attacks should mainly increase target-class inputs, whereas the *bug intuition* predicts idiosyncratic perturbations whose structure is not explained by feature representations.

Fig. 1 shows a typical AEx: an input from Class 2 (orange) is perturbed, with its IPP displayed, and is misclassified as Class 6 (brown). The perturbation does not simply increase Class 6 inputs, so it disagrees with the feature-intuition baseline. However, it is also not random: the signs and magnitudes align systematically with the configuration of latent feature directions. This visible pattern motivates the third, empirically testable intuition used below: *attacks are mediated by latent geometry*.

The IPP provides the first quantitative link between input-space changes and latent geometry. The sign and magnitude of each class-level perturbation $\delta^{(j)}$ track the alignment between that class’s latent representation $\mathbf{v}^{(j)}$ and the latent attack vector $\Delta\mathbf{h}$. Classes with positive alignment ($\mathbf{v}^{(j)} \cdot \Delta\mathbf{h} > 0$) receive amplified input features, whereas negatively aligned classes are attenuated. Thus, the input changes in Fig. 1(b) are the input-space expression of the dot products shown in Fig. 1(c), and those dot products determine movement across the decision boundary.

To test whether this pattern is specifically due to superposition geometry, we compare PGD-generated attacks with theoretically optimal perturbations derived from the interference mechanism in Sec. 3.2. For each (k, m) configuration, we train 5 random seeds, generate 1000 PGD attacks per model, and compute the cosine similarity between each successful perturbation and its corresponding optimum. The baseline is norm-matched random perturbations, so the comparison isolates directional agreement rather than perturbation size. Tab. 1 shows consistently high PGD–theory alignment across dimensionalities: 0.97 ± 0.02 for $(6, 2)$, 0.96 ± 0.00 for $(30, 10)$, and 0.92 ± 0.00 for $(90, 30)$, while the random baseline remains at 0.00 up to small standard deviations. We filter out samples with ℓ_2 norm less than ϵ to ensure that the comparison uses meaningful perturbations. One-sample t-tests against the random baseline give $p < 10^{-10}$ in all configurations, supporting the conclusion that PGD attacks match the interference-based optimum rather than chance directions.

Finding: Adversarial attacks systematically exploit interference between superposed features. Across configurations, successful PGD attacks align with the geometry-derived optimum and separate sharply from norm-matched random baselines.

Do input correlations determine latent feature geometry? We next ask whether the learned geometry itself is stable across models or varies with the input distribution. We manipulate the correlation structure in the training data to create three ordered conditions with increasing constraint strength. In the *uncorrelated* condition, features are sampled independently as before. In the *paired* condition, each feature in class i is coupled with a feature in class j , so the two activate together. In the *global* condition, each sample has a random phase that creates a cyclic activation pattern: adjacent classes have similar activation probabilities and frequently co-activate, whereas distant classes co-activate rarely.

Fig. 2 shows how these correlation conditions change the arrangement of $\mathbf{v}^{(j)}$. We quantify geometric similarity by comparing the pairwise cosine-similarity matrices between all feature pairs in each model and then correlating those matrices across random seeds; higher values mean more similar learned geometries. For each condition, we evaluate 25 seed pairs. In the $m = 2, k = 7$ setup, geometric similarity increases from 0.23 ± 0.14 (uncorrelated) to 0.47 ± 0.32 (paired) and

Table 1: PGD-discovered attacks closely match theoretically optimal perturbations and clearly separate from the random baseline across (k, m) configurations. Values are mean $\pm$ std over 1000 attacks per condition; all comparisons against the random baseline have p -values below 10^{-10} .

k	m	Cosine Sim. (PGD vs Theory)	Cosine Sim. (Random Baseline)
6	2	0.97 ± 0.02	0.00 ± 0.02
30	10	0.96 ± 0.00	0.00 ± 0.01
90	30	0.92 ± 0.00	0.00 ± 0.00

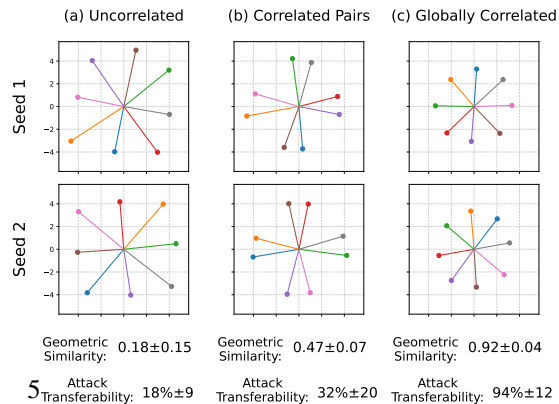


Figure 2: Input correlations increase geometric consistency across initialisations and correspondingly

0.88 $\pm$ 0.06 (global) (mean $\pm$ std). This monotonic pattern makes the margin between conditions explicit: paired correlations partially constrain the arrangement relative to uncorrelated data, while global correlations produce near-identical geometries across seeds. Pairwise two-sample t-tests with Bonferroni correction give $p < 10^{-3}$ for all three comparisons ($\alpha = 0.05$ corrected for 3 comparisons). Results across additional (k, m) configurations are reported in Sec. C.3.

The mechanism behind the trend is *interference avoidance*: features that co-activate frequently are arranged to reduce mutual interference. Stronger correlations therefore reduce the degrees of freedom available to the learned representation. Uncorrelated data allows many geometries that separate classes, paired correlations fix the relative positions of coupled features, and global correlations determine an essentially unique geometry up to rotation.

Finding: Input correlations constrain feature geometry. As correlation strength increases, geometric similarity rises across seeds, showing that stronger data constraints reduce representational degrees of freedom.

Does shared geometry explain attack transferability? Having established that input correlations make latent geometries more consistent across initialisations, we test whether this same consistency explains attack transferability. The prediction is comparative: if perturbations are tuned to a particular interference pattern, transfer rates should be highest between models with the most similar geometric arrangements and lower when geometry varies across seeds.

Using the same seed pairs, we generate adversarial attacks on source models and measure success when those attacks are applied to target models within each correlation condition, giving 25 transfer measurements per condition. All models have $> 95\%$ clean accuracy, so transfer differences are not driven by clean-performance failures. In the $k = 7, m = 2$ configuration (Fig. 2), transfer follows the same ordering as geometric similarity: globally correlated data gives 94% $\pm$ 12% attack transfer (mean $\pm$ std), paired correlations give 32% $\pm$ 20%, and uncorrelated data gives only 18% $\pm$ 9%.

This pattern is practically significant because each perturbation component amplifies or suppresses features to create constructive interference across a decision boundary. When the same perturbation is applied to a model with a different feature arrangement, the components that were constructive for the source can cancel or interfere destructively in the target, reducing transfer success.

Finding: Attack transferability is governed by shared interference patterns: the highest transfer rates occur in the condition with the most consistent latent geometry.

Does reducing superposition suppress these attacks? If superposition creates vulnerability through interference, reducing superposition should improve the relevant robustness metrics or redirect the attack budget away from orthogonal features. We test this prediction in three ways. First, we set $m = k$, allowing the network to learn orthogonal class directions $\mathbf{v}^{(j)}$. In this setting, a perturbation that changes the model’s prediction also changes the ground-truth class: moving a sample from class A to class B requires making the class-B feature sum exceed the class-A sum, genuinely transforming the sample. Across 1000 attempts at every tested ϵ , we find zero successful adversarial examples. Second, we isolate superposition pressure from capacity by fixing m and varying k ; robust accuracy decreases monotonically with k/m (App. Tab. 5), showing that vulnerability scales with the degree of interference. Third, we constrain one class’s latent vector $\mathbf{v}^{(\perp)}$ to remain orthogonal to all others during training. For attacks between classes that remain in superposition (e.g. from class a to b where

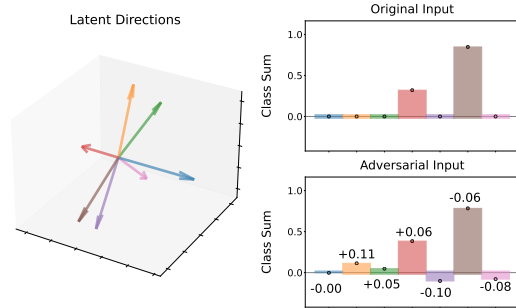


Figure 3: When one class is represented orthogonally, an adversarial attack from class 5 to class 3 spends no visible perturbation budget on that orthogonal class (class 1), isolating the budget on superposed features.

$\mathbf{v}_a \not\perp \mathbf{v}_b$), the inputs for the orthogonal class remain unperturbed (Fig. 3), showing that the attack budget concentrates on features participating in interference.

Finding: Adversarial attacks allocate their perturbation budget to features in superposition, while orthogonal features remain comparatively unused.

3.2 FORMAL ANALYSIS

We now formalise the mechanism behind the empirical comparisons. In the linear setting, the relationship between superposition interference and adversarial vulnerability can be written exactly, turning the observed PGD–theory alignment into a prediction rather than a post-hoc correlation. We use the notation from the experimental setup: $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ induces superposition through columns $\mathbf{v}_i$, the decoder corresponds to the encoder transposed as empirically observed, and the logit for class j is $z_j = \mathbf{v}_j^\top \mathbf{h}$.

Proposition 1. *The optimal input perturbations δ that maximise movement from class j to class k under constraint $\|\delta\|_2 = \epsilon$ satisfy $\delta \propto \mathbf{W}_e^\top \mathbf{n}$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to the decision boundary between classes.*

Proof Sketch. To move from class j to k , we maximise the logit margin $z_k - z_j = \mathbf{n}^\top (\mathbf{h} + \Delta \mathbf{h})$, where $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ and $\Delta \mathbf{h} = \mathbf{W}_e \delta$. This becomes $\max_{\|\delta\|_2 = \epsilon} \delta^\top \mathbf{W}_e^\top \mathbf{n}$. By Cauchy-Schwarz, this is maximised when $\delta \propto \mathbf{W}_e^\top \mathbf{n}$. Full proof in Sec. B. $\square$

Corollary 1 (Interference Drives Vulnerability) *The adversarial perturbation magnitude for feature i is $|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$, directly proportional to the differential interference between feature i and the class representations.*

This expression makes the quantitative role of superposition explicit. Each input feature i receives perturbation magnitude proportional to its differential interference with the source and target class representations. When features are non-orthogonal, even semantically unrelated inputs can affect the class decision; adversarial perturbations exploit exactly these measured dependencies to change the output.

Proposition 2. *Models with feature representations related by orthogonal transformation $\mathbf{Q}$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$) share identical optimal attack directions in input space.*

Proof Sketch. Under transformation $\mathbf{v}'_i = \mathbf{Q} \mathbf{v}_i$, optimal perturbations remain invariant: $\delta'_i \propto (\mathbf{v}'_i)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = \mathbf{v}_i^\top \mathbf{Q}^\top \mathbf{Q} (\mathbf{v}_k - \mathbf{v}_j) = \mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j) = \delta_i$. Full proof in Sec. B. $\square$

Together with the empirical results, these propositions give a **mechanistic pathway** whose quantities can be compared at each step. Specifically: (i) input correlations constrain feature arrangements in superposition, (ii) those arrangements determine interference patterns, (iii) the interference patterns specify optimal perturbations via $\delta \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$ (Proposition 1), and (iv) shared geometric constraints produce similar interference patterns across models, increasing transferability (Proposition 2):

Correlations $\xrightarrow{\text{constrain}}$ **Feature Geometry** $\xrightarrow{\text{determines}}$ **Interference Patterns** $\xrightarrow{\text{enable}}$ **Transferability**

4 ATTACKS IN REALISTIC MODELS

We next test whether the toy-model comparison pattern appears in a more realistic vision setting: a ViT (Dosovitskiy et al., 2020) trained on CIFAR-10 (Krizhevsky, 2009) (with a ResNet-18 comparison in Sec. D.2.1 (He et al., 2016)). The intervention is deliberately matched to the toy setup: we add a bottleneck layer immediately before the final classifier so that the $k = 10$ class representations must be compressed into an m -dimensional space. This lets us compare three quantities as superposition pressure changes: normalised robust accuracy, attack transferability across seeds, and consistency of the learned class geometry.

Setup. We train the ViT architecture (6 layers, patch size 4, residual stream $d = 512$) on CIFAR-10 across five random seeds. After training, we replace the original classification head with a bottleneck composed of a linear encoder that compresses the pre-classification activations to m dimensions and

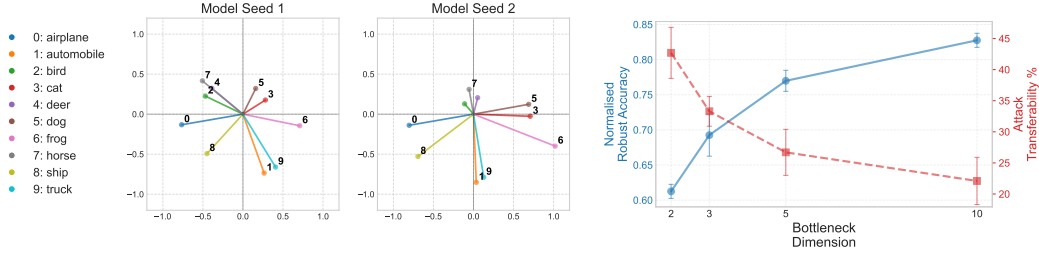


Figure 4: CIFAR-10 bottleneck experiments show the same comparison pattern as the toy setting. Left: class representations retain similar ordering and clustering across seeds. Right: as bottleneck dimension m increases, superposition pressure decreases, attack transferability falls, and normalised robust accuracy rises.

a linear decoder that maps back to the $k = 10$ CIFAR-10 classes. The bottleneck is trained with CE loss and no ReLU, matching the main toy-model setting in Sec. 3; during this stage, the base ViT weights are frozen. We vary $m \in \{2, 3, 5, 10\}$, so smaller m gives a stronger compression ratio and greater superposition pressure. For each bottlenecked model, we generate AExs using both ℓ_∞ - and ℓ_2 -norm PGD. We use a small step size and many iterations, stopping once the attack changes the model prediction (full details in Sec. D.1). Robust accuracy is normalised by each model’s clean accuracy, ensuring that robustness comparisons are not confounded by the lower clean accuracy of smaller bottlenecks. Attack transferability is measured as in Sec. 3.

Results. The base models have mean clean accuracy of 81%. The bottlenecked models show the expected clean-accuracy cost of compression: 54%, 67%, 76%, and 81% for $m = \{2, 3, 5, 10\}$, respectively. We therefore interpret robustness results using normalised robust accuracy, so the reported trend reflects vulnerability relative to each bottleneck model’s own clean performance.

The CIFAR-10 results reproduce the key quantitative pattern from the toy setting. First, Fig. 4 (right) shows that decreasing m – increasing superposition pressure – is associated with lower normalised robust accuracy and higher attack transferability across independently initialised models. This paired movement is the important comparison: denser packing of class representations makes small perturbations more likely to induce broad interference, while the restricted bottleneck gives different seeds fewer viable geometries, increasing shared interference patterns and thus transfer. Results across perturbation magnitudes are reported in Sec. D.1.1.

Second, Fig. 4 (left) shows that the m -dimensional class representations in the bottleneck have consistent ordering and clustering across seeds. This mirrors the toy-model result that data correlations constrain superposed geometry, and the observed clusters often match semantic class similarities, such as ‘cat’ and ‘dog’ appearing near one another. Third, when AExs generated for a bottlenecked model are passed through the corresponding base ViT, they produce similar relative logit changes (*i.e.* $\text{logit}_{\text{base}}(\mathbf{x}_{\text{adv}}) - \text{logit}_{\text{base}}(\mathbf{x}_{\text{orig}})$). The perturbations therefore affect the base and bottlenecked models in a consistent direction; the bottleneck increases the practical impact of those perturbations by making the compressed class geometry easier to push across decision boundaries.

5 ADVERSARIAL VULNERABILITY VIA ALGORITHMIC BRITTLINESS

The previous sections show that feature superposition can *suffice* for adversarial vulnerability. We now use modular addition as a contrast case to test necessity. In this setting, the relevant features are represented orthogonally rather than in superposition, yet the model remains attackable. The comparison broadens the quantitative framing of the paper: representation geometry is still predictive of vulnerability, but the measured failure mode is algorithmic brittleness rather than superposition interference. We ask three questions: *how do such attacks work mechanistically, can this understanding construct attacks directly without gradients, and does naive robustness certification change the failure mode or mainly increase the required attack budget?*

Setup. We train a linear encoder + MLP on one-hot integers (a, b) to predict $(a + b) \bmod P$. As detailed in Sec. E and Nanda et al. (2023), the model learns a trigonometric algorithm for the

task. As part of this learned algorithm¹ the encoder projects inputs onto non-basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at key frequencies $\omega_k = 2\pi k/P$. The MLP then composes $\cos((a+b)\omega_k)$ through sum-angle identities and reaches 100% accuracy. Crucially for the comparison with earlier sections, these features are represented *orthogonally* (i.e. not in superposition). We generate AExs by relaxing the one-hot constraint and perturbing only input a with PGD, thereby moving inputs off the training manifold while keeping the learned frequency basis fixed.

Attacks exploit the learned key frequencies. Fourier analysis of successful perturbations δ (App. Fig. 6) shows that their dominant frequencies match the learned algorithm’s key frequencies, ω_k . This frequency specificity is also visible in transfer: 0% of attacks transfer to models with different ω_k , such as models trained from different seeds. The frequency-aligned perturbations cause misclassification by altering the sinusoidal values read from the encoder features. A perturbation δ turns the one-hot $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$, and the newly non-zero inputs add contributions that corrupt the sinusoidal values used by the downstream MLP. Because the learned algorithm relies on constructive interference of each ω_k at the logits, changing a single ω_k is not enough; successful attacks must perturb multiple frequency components in a coordinated way. We give the detailed mechanistic intuition in Sec. E.1.

Analytic construction of attacks. The frequency mechanism supports a direct quantitative attack construction. We build gradient-free, *informed* perturbations from composite sinusoidal waves at matching ω_k . On $0 + 0 \bmod P$, these attacks succeed with $\|\delta\|_2=0.22$, close to PGD (0.17, $\approx 1.3\times$ the PGD budget) and far below naive alternatives: single-index perturbations require 0.97, ℓ_∞ noise requires 2.04, and uniform perturbations require 6.20. These margins show that the representation-aware construction targets the relevant mechanism rather than merely producing a post-hoc explanation.

Do defences mitigate this algorithmic vulnerability? We apply certified training against ℓ_∞ perturbations (Müller et al., 2023). At fixed attack budgets, the *robust* accuracy improves from pre-robustified $\rightarrow$ post-robustified as follows: 0% $\rightarrow$ 85% at $\epsilon_\infty = 5e-3$, 0% $\rightarrow$ 96% at $2e-3$, and 76% $\rightarrow$ 100% at $1e-3$. These are substantial gains under the evaluated budgets. However, when attacked with PGD, the robust model still fails through the same frequency-based brittleness, although it requires larger perturbation norms. Certification therefore improves budget-level robustness without eliminating the identified semantic fragility, motivating defences that target the specific frequency manipulation rather than relying only on broad norm bounds.

Relation to the paper’s thesis. This contrast case makes two comparison points precise. First, superposition is *sufficient but not necessary*: orthogonal representations can still be vulnerable when the learned algorithm is brittle. Second, analysing AExs through the lens of *architecture–semantics interaction* is operationally useful because it yields both representation-informed perturbations and a clearer target for defences.

Although this mechanism is less central than superposition interference, it shares an important quantitative dependence on sparsity assumptions: upstream violations of sparsity (e.g. noise or superposed features) can deliver effectively ‘pre-perturbed’ inputs to downstream orthogonal circuits, activating the same frequency-based failure.

6 RELATED WORK

Superposition & latent representations. Work on superposition provides the representational quantities that we compare against adversarial behaviour. Nanda (2023) distinguish *representational* from *computational* superposition, while Gurnee et al. (2023) separate *alternating* and *simultaneous* interference. Prior results also show that data correlations shape the geometry of learned representations: correlated feature pairs become orthogonal (Elhage et al., 2022), correlations can drive superposition formation (Chan, 2024), and semantic clustering can emerge from correlation structure (Prieto et al., 2025a). More broadly, representation geometry is influenced by spectral bias (Rahaman et al., 2019), neural collapse (Kothapalli, 2023), and optimisation objectives (Casper, 2023). Sparse autoencoders (SAEs) have also been applied to vision models to disentangle features in superposition (Lim et al., 2024; Joseph et al., 2025). Our contribution is to place these geometric

¹We refer to the specific computation executed by the network to solve the task as the *model’s learned algorithm*. As this algorithm depends on extracting precise values from learned feature representations (frequencies), it remains *brittle* in the face of small perturbations to the bases. We term this the *algorithmic brittleness*.

measurements directly beside adversarial metrics such as perturbation alignment, robust accuracy, and transferability.

Adversarial vulnerability. The adversarial robustness literature is often organised around attack–defence comparisons. Attacks (Szegedy et al., 2014) use small perturbations to maximise model error, with FGSM (Goodfellow et al., 2014) and PGD (Madry et al., 2018) serving as standard baselines. Robustness then measures how much performance is retained under these perturbation budgets. Defences include adversarial training (Madry et al., 2018) and certified training (Wong & Kolter, 2018; Palma et al., 2024), which provide guarantees within specified input bounds. Our framing complements this work by asking which learned representational quantities explain the observed attack success and transfer patterns under the same budgeted evaluations.

The role of input & latent features on adversarial robustness. Several accounts attribute vulnerability to predictive but brittle ‘non-robust’ input features (Ilyas et al., 2019; Goh, 2019) or learning shortcuts (Li et al., 2023), and perturbation characteristics are known to vary by dataset (Maiya et al., 2021). Elhage et al. (2022) suggest a possible link between superposition and AExs, while Casper (2023) debates this connection. Empirical work has also observed a tension between representational efficiency and adversarial robustness (Barsbey et al., 2025); Zhang et al. (2021) show that adversarial perturbations push latent representations across decision boundaries; and Ganeshan et al. (2019) find that PGD targets final layers and propose feature-based attacks. Transferability is limited by representation discrepancies between models (Li et al., 2023; Wang et al., 2024), and decorrelating features reduces transfer (Wiedeman & Wang, 2022). These findings align with our evidence framing: the strongest transfer occurs when models share the latent interference patterns that make a perturbation effective.

7 DISCUSSION & CONCLUDING REMARKS

We have shown that adversarial attacks can exploit interference patterns arising from the geometry of superposed features in NNs. Across controlled toy models and bottlenecked ViTs, the evidence follows a consistent comparison structure: stronger superposition produces more exploitable interference, PGD perturbations align with geometry-derived optima rather than random baselines, input correlations make learned geometries more similar across seeds, and shared geometry increases attack transferability. The practical implication is that adversarial vulnerability can be quantitatively predictable from how networks compress information, rather than viewed only as an irregularity of optimisation or a property of non-robust inputs.

Limitations & future work. The strongest evidence in this paper comes from simplified and deliberately controlled models. These settings make the relevant quantities measurable – feature directions, bottleneck dimension, perturbation alignment, robust accuracy, and transfer – but they do not cover the full range of attacks and architectures studied in adversarial robustness. For example, constrained attack types (*e.g.* universal perturbations, jailbreaks, patch attacks) may not be able to activate the precise feature combinations required to exploit superposition interference, and large-scale models may distribute relevant features across layers rather than a single engineered bottleneck.

Future work should quantify the ‘adversarial cost’ of superposition more directly, connect that cost to accuracy–robustness trade-offs, and measure how robust training reshapes superposition geometry. Our current experiments use simplified settings with class features placed in engineered superposition, whereas real networks contain unknown, unlabelled features that interfere across many layers. We plan to use SAEs (Bricken et al., 2023) to tackle these challenges in large-scale networks, with preliminary results in Sec. F suggesting that this is a promising direction. The central empirical target remains the same: compare attack behaviour against measured representation geometry while preserving the distinction between superposition-driven interference and other forms of algorithmic brittleness.

ETHICS STATEMENT

This work studies adversarial vulnerability and therefore includes attack constructions, but the empirical emphasis is diagnostic: the experiments compare perturbations against learned representation geometry to better understand when models fail. The same quantitative framing can support more targeted robustness evaluations and defences by identifying the features, frequencies, or interference patterns that attacks exploit.

REPRODUCIBILITY STATEMENT

We provide code for all experiments, including the toy-model comparisons in Sec. 3, CIFAR-10 bottleneck evaluations in Sec. 4, and modular-addition analyses in Sec. 5. The reported metrics specify the method, baseline, perturbation norm, dataset, seed structure, and evaluation condition wherever possible; architecture and training details are provided in Sec. D.1 and Sec. E.2, with additional ResNet-18 and SAE experiments documented in Sec. D.2.1 and Sec. F.

LLM USAGE STATEMENT

LLMs were used to assist with writing and editing. The scientific claims, experimental setup, numerical results, comparisons, and conclusions remain the responsibility of the authors.

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APPENDIX

A GENERATION OF ADVERSARIAL EXAMPLES

We use PGD to generate AExs because it provides a standard, budget-controlled baseline for comparing attack success across models and perturbation norms (Madry et al., 2018). For untargeted attacks, PGD maximises the classifier’s loss $\mathcal{L}(f(\mathbf{x}'), y_{\text{true}})$; for targeted attacks, it minimises $\mathcal{L}(f(\mathbf{x}'), y_{\text{target}})$ using the update rule:

$$\mathbf{x}'^{(k+1)} = \Pi_{\mathcal{S}} \left(\mathbf{x}'^{(k)} \pm \alpha \mathbf{g}_k \right) \quad (1)$$

(+ for maximisation, – for minimisation). Here $\mathbf{x}'^{(0)}$ is an initial perturbed input, α is the step size, $\mathbf{g}_k$ is the (normalised) gradient $\nabla_{\mathbf{x}} \mathcal{L}(f(\mathbf{x}'^{(k)}), y_{\text{class}})$ (y_{class} being y_{true} or y_{target}), and $\Pi_{\mathcal{S}}$ projects onto the attack budget $\mathcal{S} = \{\mathbf{x}' \mid \|\mathbf{x}' - \mathbf{x}\|_p \leq \epsilon\}$ with input-domain clipping. We report results under the following ℓ_p constraints:

1. ℓ_{∞} : Constraint $\|\delta\|_{\ell_{\infty}} = \max_i |\delta_i| \leq \epsilon$. $\mathbf{g}_k$ is $\text{sign}(\nabla_{\mathbf{x}} \mathcal{L}(\cdot))$. $\Pi_{\mathcal{S}}$ clips each $\mathbf{x}'_i$ to $[\mathbf{x}_i - \epsilon, \mathbf{x}_i + \epsilon]$.
2. ℓ_2 : Constraint $\|\delta\|_{\ell_2} = \sqrt{\sum_i \delta_i^2} \leq \epsilon$. $\mathbf{g}_k$ is $\nabla_{\mathbf{x}} \mathcal{L}(\cdot) / \|\nabla_{\mathbf{x}} \mathcal{L}(\cdot)\|_{\ell_2}$. $\Pi_{\mathcal{S}}$ rescales $\delta = \mathbf{x}' - \mathbf{x}$ if $\|\delta\|_{\ell_2} > \epsilon$ via $\delta \leftarrow \epsilon \cdot \delta / \|\delta\|_{\ell_2}$.

B OMITTED PROOFS & THEORETICAL RESULTS

To connect the empirical alignment results to a precise mechanism, we prove that optimal perturbations weight each input dimension according to how strongly its corresponding feature aligns with movement toward the decision boundary. We analyse the linear versions of our models without ReLU activation functions so that the relevant comparison can be written exactly.

Consider an input $\mathbf{x} \in \mathbb{R}^d$ encoded as $\phi(\mathbf{x}) = \mathbf{W}_e \mathbf{x}$ into a latent representation $\mathbf{h} \in \mathbb{R}^m$, where the columns of $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ form an overcomplete dictionary $\{\mathbf{v}_i\}_{i=1}^d$ with $m < d$. In the argmax task, we empirically observe that the trained decoder is the transpose of the encoder, $\mathbf{W}_d = \mathbf{W}_e^{\top}$. The logit for class j is therefore $z_j = \mathbf{v}_j^{\top} \mathbf{h}$, where $\mathbf{v}_j$ is the j -th column of $\mathbf{W}_e$, and the predicted class is $\hat{y} = \arg \max_j z_j$. Input-space perturbations map to latent perturbations by $\Delta \mathbf{h} = \mathbf{W}_e \delta$, with $\delta = (\delta_1, \dots, \delta_d)^{\top}$ giving the perturbation coefficients.

The decision boundary between classes j and k is the set of points with equal logits, $\mathcal{B}_{jk} = \{\mathbf{h} : z_j = z_k\}$, equivalently $\{\mathbf{h} : (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} = 0\}$. Thus $\mathbf{n} = (\mathbf{v}_k - \mathbf{v}_j)$ is the normal to the pairwise boundary and the direction that maximally increases the margin $z_k - z_j$. We recall the propositions before giving the proofs, emphasising the same margin quantity used in the empirical attack comparisons.

Proposition 1 (Optimal targeted attack). The optimal input perturbations δ that maximise movement from class j toward class k under constraint $\|\delta\|_2 = \epsilon$ satisfy:

$$\delta \propto \mathbf{W}_e^{\top} \mathbf{n}$$

Proof. To move a sample from being classified as j to being classified as k , we need to maximise the logit margin $z_k - z_j$. Under perturbation δ , the new margin becomes:

$$z'_k - z'_j = \mathbf{v}_k^{\top} (\mathbf{h} + \Delta \mathbf{h}) - \mathbf{v}_j^{\top} (\mathbf{h} + \Delta \mathbf{h}) \quad (2)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} (\mathbf{h} + \mathbf{W}_e \delta) \quad (3)$$

$$= (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{h} + (\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta \quad (4)$$

The change in margin is $(\mathbf{v}_k - \mathbf{v}_j)^{\top} \mathbf{W}_e \delta = \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. We seek:

$$\max_{\delta} \delta^{\top} \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j) \quad \text{subject to} \quad \|\delta\|_2 = \epsilon$$

Let $\mathbf{g} = \mathbf{W}_e^{\top} (\mathbf{v}_k - \mathbf{v}_j)$. By the Cauchy-Schwarz inequality:

$$|\delta^{\top} \mathbf{g}| \leq \|\delta\|_2 \|\mathbf{g}\|_2 = \epsilon \|\mathbf{g}\|_2$$

Equality is achieved when δ and $\mathbf{g}$ are parallel. Given the constraint $\|\delta\|_2 = \epsilon$:

$$\delta = \frac{\epsilon}{\|\mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)\|_2} \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

□

Corollary (Interference drives vulnerability). For a targeted attack from class j to class k , the adversarial perturbation magnitude for input feature i is:

$$|\delta_i| \propto |\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)|$$

where $\mathbf{v}_i$ is the i -th column of $\mathbf{W}_e$. This quantity represents the differential interference between feature i and the class representations.

This reveals the mechanism by which superposition creates adversarial vulnerability. Each input feature i is perturbed proportionally to $\mathbf{v}_i^\top (\mathbf{v}_k - \mathbf{v}_j)$ —the differential interference between feature i and the class representations. Under superposition, the non-orthogonality means that even semantically unrelated features have non-zero inner products with $(\mathbf{v}_k - \mathbf{v}_j)$, creating exploitable interference patterns. The multi-class setting amplifies this vulnerability, as with k classes there are multiple possible pairwise boundaries, each creating a distinct interference pattern. Adversarial perturbations leverage these cross-feature dependencies—they manipulate features that affect it through their interference with the class representations. This explains the vulnerability we observe empirically: attacks succeed not by directly increasing target class features, but by exploiting the web of interference created by superposition.

Prop. 1 characterises the optimal perturbation direction for moving from class j to k , providing the gradient for maximising the margin $z_k - z_j$. When intervening classes exist (where $z_j < z_i < z_k$ for some class i), following this gradient might cause the model to predict i before reaching target k . Iterative methods like PGD handle this by recomputing gradients at each step—the global attack path emerges from repeated local decisions rather than a single optimisation. Furthermore, while our analysis assumes $\mathbf{W}_d = \mathbf{W}_e^\top$ based on our empirical observations, the framework extends to arbitrary decoders. In the general case, with decoder $\mathbf{W}_d \in \mathbb{R}^{k \times m}$ having rows $\mathbf{w}_j^\top$, the optimal perturbation becomes $\delta \propto \mathbf{W}_e^\top (\mathbf{w}_k - \mathbf{w}_j)$, capturing interference between encoder directions and decoder weights.

Proposition 2 (Attack transferability). Consider encoders ϕ and ψ with basis matrices $\mathbf{W}_e \in \mathbb{R}^{m \times d}$ and $\mathbf{W}'_e \in \mathbb{R}^{m \times d}$ whose columns are related by orthogonal transformation $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$ (where $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$). If both models use their encoder transpose as decoder (i.e., $\mathbf{W}_d = \mathbf{W}_e^\top$ and $\mathbf{W}'_d = (\mathbf{W}'_e)^\top$), then both models have identical optimal input perturbation vectors for any targeted attack from class j to class k .

Proof. For model ϕ , the optimal perturbation from Prop. 1 is:

$$\delta^\phi \propto \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j)$$

For model ψ with transformed columns $\mathbf{v}'_i = \mathbf{Q}\mathbf{v}_i$:

$$\delta^\psi \propto (\mathbf{W}'_e)^\top (\mathbf{v}'_k - \mathbf{v}'_j) = (\mathbf{W}'_e)^\top (\mathbf{Q}\mathbf{v}_k - \mathbf{Q}\mathbf{v}_j)$$

Since $\mathbf{W}'_e = \mathbf{Q}\mathbf{W}_e$ (all columns are transformed), we have:

$$\delta^\psi \propto (\mathbf{Q}\mathbf{W}_e)^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{5}$$

$$= \mathbf{W}_e^\top \mathbf{Q}^\top \mathbf{Q}(\mathbf{v}_k - \mathbf{v}_j) \tag{6}$$

$$= \mathbf{W}_e^\top (\mathbf{v}_k - \mathbf{v}_j) \tag{7}$$

where the last equality uses $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$. Thus δ^ϕ and δ^ψ have identical proportionality. Under the same norm constraint $\|\delta\|_2 = \epsilon$, we have $\delta^\phi = \delta^\psi$. □

Prop. 2 is used to explain why attacks transfer between models with similar training regimes. When models learn feature representations that differ only by orthogonal transformation—essentially the

same geometric structure in different orientations—they share identical vulnerability patterns in input space. The orthogonal transformation preserves all inner products between features, maintaining the interference patterns that attacks exploit. This provides an explanation for our empirical observation that models trained on data with the same correlations exhibit high attack transferability: they discover similar feature geometries up to rotation, leading to shared adversarial vulnerabilities.

C TOY MODEL EXPERIMENTS

This section provides supplementary details and extended quantitative results for the toy-model experiments in Sec. 3. We report model accuracies across a wider range of parameters to show how bottleneck capacity, class count, feature count, and sparsity affect the conditions under which feature superposition is learned. We then add visual examples corresponding to Fig. 1, emphasising how input perturbation profiles, latent-space movement, and attack norms compare across conditions.

C.1 HYPOTHESES TESTING FRAMEWORK

We explicitly state our hypotheses for the three research questions in Sec. 3.

Research Q1: Do adversarial perturbations exploit superposition geometry?

- H_0 : Adversarial perturbations are random with respect to feature geometry.
- H_1 : Adversarial perturbations systematically exploit geometric relationships between superposed representations.

Research Q2: Do data correlations determine superposition geometry?

- H_0 : Input correlations have no systematic effect on learned geometries.
- H_1 : Input correlations determine geometric arrangements across model initializations.

Research Q3: Does shared geometry explain attack transferability?

- H_0 : Attack transferability is independent of geometric similarity.
- H_1 : Transferability increases with shared latent structure.

We test these hypotheses through controlled experiments:

- $H_1(1)$: We measure the input perturbation profile alignment with a class’s latent representation and the latent attack vector.
- $H_1(2)$: We systematically vary input correlations and measure resulting geometries.
- $H_1(3)$: We quantify transferability rates across models with varying geometric similarity.

C.2 ACCURACY OF TOY MODEL FOR A RANGE OF PARAMETERS

The main text in Sec. 3 uses low-dimensional toy settings so that the geometry and attack profiles are visually interpretable. This subsection broadens the quantitative context by reporting CE toy-model accuracy across hidden layer size (h), number of classes (k), total features, and sparsity (S). Sparsity is the probability that an input feature $x_i^{(j)}$ is set to zero, so feature density $1 - S$ is the probability that a feature is non-zero. Tab. 2 and Tab. 3 show where the model retains high classification accuracy despite compression and where increasing class count or feature density makes the superposition problem harder.

C.3 CORRELATIONS CONSTRAIN ARRANGEMENTS

Tab. 4 extends the main-text test of whether input correlations determine latent feature geometry across additional class counts (k) and hidden dimensions (m). The table keeps the metric fixed – geometric similarity across seeds – so the comparison isolates how correlation type changes representational consistency.

Table 2: Classification accuracy of the CE toy model at fixed bottleneck dimension ($m = 2$), comparing numbers of classes (k), total input features (features = $k \times 3$), and input feature densities ($1 - S$). The table shows how higher sparsity helps preserve accuracy as more classes are compressed into the same latent space.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
3	9	2	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
4	12	2	0.67	0.71	0.82	0.92	0.98	0.99	1.00	1.00
5	15	2	0.53	0.50	0.65	0.77	0.89	0.95	0.98	0.99
6	18	2	0.40	0.43	0.51	0.66	0.82	0.93	0.97	0.99
7	21	2	0.34	0.34	0.40	0.53	0.73	0.87	0.95	0.98
8	24	2	0.30	0.30	0.33	0.40	0.63	0.82	0.93	0.97
9	27	2	0.24	0.26	0.30	0.35	0.57	0.75	0.89	0.96
10	30	2	0.22	0.24	0.26	0.31	0.50	0.72	0.87	0.95
15	45	2	0.14	0.15	0.16	0.17	0.25	0.40	0.65	0.86
20	60	2	0.10	0.10	0.11	0.13	0.15	0.24	0.44	0.76
25	75	2	0.07	0.08	0.09	0.10	0.12	0.16	0.26	0.62
30	90	2	0.06	0.07	0.07	0.07	0.09	0.11	0.21	0.45

Table 3: Classification accuracy of the CE toy model across class counts (k), total input features, bottleneck dimensions (m), and sparsity levels (S), including settings with different numbers of features per class (p). These comparisons show how capacity and sparsity jointly affect performance under superposition.

Classes (k)	Features	Hidden (m)	Accuracy at Input Feature Density ($1 - S$)							
			1.0	0.57	0.33	0.19	0.11	0.06	0.04	0.02
30	30	30	0.23	0.24	0.38	0.62	0.83	0.94	0.99	1.00
30	90	90	0.27	0.27	0.33	0.41	0.51	0.67	0.85	0.94
40	40	30	0.67	0.54	0.73	0.77	0.88	0.96	0.99	0.99
40	120	30	0.71	0.64	0.65	0.72	0.73	0.79	0.89	0.96
60	60	10	0.05	0.07	0.12	0.25	0.47	0.73	0.90	0.97
60	180	10	0.08	0.10	0.13	0.17	0.22	0.32	0.52	0.75
80	80	30	0.15	0.17	0.23	0.41	0.63	0.79	0.91	0.98
80	240	30	0.23	0.22	0.31	0.41	0.48	0.53	0.66	0.80
100	100	10	0.03	0.04	0.05	0.10	0.21	0.43	0.69	0.87
100	300	10	0.04	0.05	0.06	0.09	0.12	0.15	0.25	0.45

Table 4: Geometric similarity across correlation types and superposition configurations. Global correlations consistently produce the highest cross-seed similarity, paired correlations are intermediate, and uncorrelated inputs yield the least constrained geometries.

Correlation Type	k	m	Geometric Similarity
Uncorrelated	6	2	0.18 ± 0.15
	30	10	0.17 ± 0.02
	90	30	0.17 ± 0.01
Paired	6	2	0.47 ± 0.07
	30	10	0.26 ± 0.07
	90	30	0.25 ± 0.01
Global	6	2	0.92 ± 0.04
	30	10	0.88 ± 0.01
	90	30	0.80 ± 0.01

C.4 HOW THE TOY MODEL MAPS TO REAL-WORLD MODELS

Scaling changes both capacity and the number of useful features a model can represent, so its effect on adversarial vulnerability is not captured by size alone. Empirical evidence suggests that larger models can be somewhat more robust to adversarial attacks, but the effect is weak. With adversarial training, scaling trends become clearer, although the gains remain largely tied to the attack type used during training rather than providing general robustness (Howe et al., 2025).

Two quantities move together as models scale. Larger models have more capacity to represent concepts, but they also face a longer tail of useful concepts in broad tasks such as next-token prediction over internet text. Consequently, increased capacity does not remove the pressure to compress features, and superposition appears to remain prevalent even in frontier models (Lindsey et al., 2025). Dictionary-learning evidence points in the same direction: SAEs require increasingly large dictionaries for larger models (Gao et al., 2025), suggesting that the feature count itself grows with model size. The core trade-off measured in this paper – more features than cleanly separable dimensions – therefore plausibly persists at scale.

Because both superposition and adversarial vulnerability persist in large-scale models, the relevant future comparison is not whether scale removes the mechanism, but how scale changes the geometry, margins, and attack budgets associated with it. Measuring those changes could identify which forms of superposition are most costly for robustness and which are comparatively benign.

C.4.1 SEPARATING SUPERPOSITION EFFECTS FROM CAPACITY REDUCTION

To separate superposition pressure from overall capacity, we keep the bottleneck dimension fixed and vary the number of classes. Specifically, we fix $m = 2$ and vary k , so changes in k/m reflect stronger or weaker compression into the same latent dimension. Tab. 5 reports clean accuracy, robust accuracy, and attack transferability under this control. The pattern shows that increasing superposition pressure (k/m), even with capacity held fixed, drives the adversarial vulnerabilities observed in the main text.

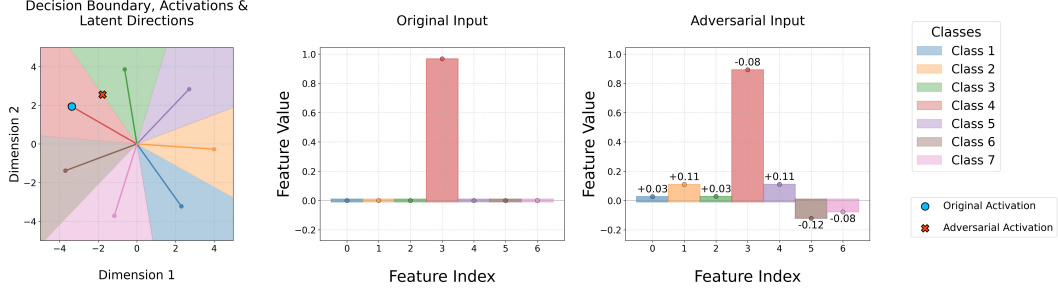
C.4.2 ADDITIONAL EXAMPLES OF AEXS IN TOY MODEL

Fig. 1 shows one representative attack whose IPP follows latent interference. Here we add visual examples in Fig. 5 to show that the same comparison between input-space perturbations and latent-space movement appears across norms and bottleneck dimensions. Specifically, we include:

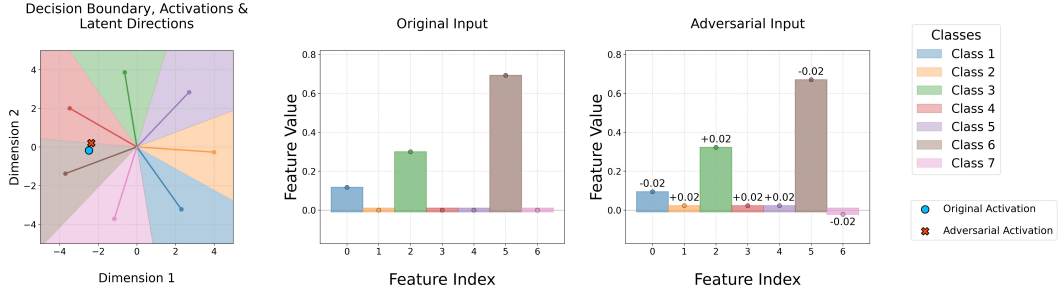
- Fig. 5a shows an additional instance of the setup in Sec. 3.1 ($m = 2, k = 7$) with an ℓ_2 -norm PGD attack, demonstrating the IPP and latent space manipulations that lead to misclassification.
- Fig. 5b shows a similar setup ($m = 2, k = 7$) using an ℓ_∞ -norm PGD attack on a less sparse input.
- Fig. 5c shows an example with increased bottleneck dimensionality ($m = 3, k = 7$) and accompanying ℓ_2 -norm PGD attack. The feature vector similarity matrix used to calculate geometric similarity is also shown.

Table 5: Clean accuracy, robust accuracy, and attack transferability across correlation types, perturbation budgets (ε), and class counts (k) with fixed bottleneck dimension. The table isolates the effect of superposition pressure from bottleneck capacity.

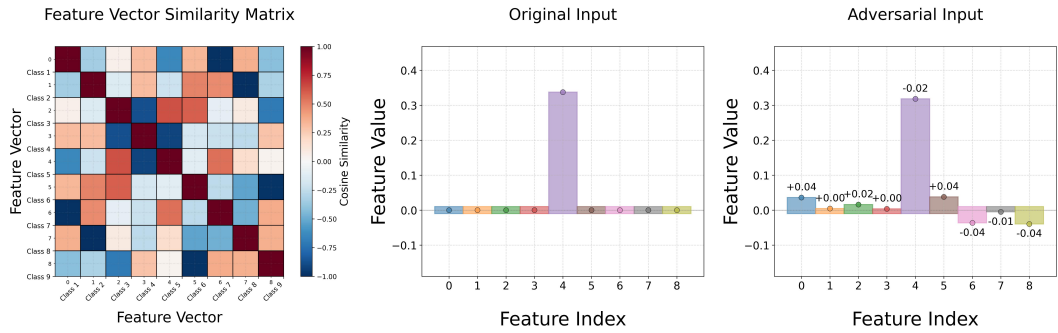
Correlation Type	Parameters		Performance (%)		
	ε	k	Accuracy	Robust Acc.	Attack Trans.
Uncorrelated	0.05	2	100.0	100.0	0.0
		4	100.0	94.9	21.2
		6	99.9	87.0	9.8
		8	99.8	77.7	17.5
	0.1	2	100.0	100.0	0.0
		4	100.0	88.1	9.8
		6	99.7	75.4	9.9
		8	99.6	56.3	23.5
	0.5	2	100.0	100.0	0.0
		4	100.0	99.4	0.0
		6	99.9	62.9	30.5
		8	99.5	24.9	39.0
	0.05	2	100.0	100.0	0.0
		4	100.0	95.8	98.4
		6	97.3	70.6	98.4
		8	93.0	57.4	97.2
Fully Correlated	0.1	2	100.0	100.0	0.0
		4	100.0	88.7	98.5
		6	97.7	45.6	99.7
		8	94.2	34.8	98.6
	0.5	2	100.0	100.0	0.0
		4	100.0	90.0	100.0
		6	97.2	34.6	100.0
		8	93.0	32.0	100.0



(a) An ℓ_2 -norm attack from class 4 to class 3, comparing original and adversarial activations in latent space with the corresponding original and perturbed input feature values.



(b) An ℓ_∞ -norm attack from class 6 to class 4, showing how the input perturbation profile corresponds to movement relative to class latent directions.



(c) An ℓ_2 -norm attack in a 7-class setup with bottleneck dimension $m = 3$, including the pairwise cosine-similarity matrix used to quantify latent representation geometry.

Figure 5: Additional toy-model AExs showing the same evidence structure as the main figure: perturbation profiles in input space align with movement through activation space under varied norms and bottleneck dimensions.

D CIFAR-10 EXPERIMENTS

Sec. 4 tests whether the toy-model evidence pattern extends to CIFAR-10 (Krizhevsky, 2009) with an engineered bottleneck. This appendix gives the architecture, training details, and extended tables used to compare bottleneck dimension, perturbation magnitude, normalised robust accuracy, and attack transferability for the experiments in Sec. 4.

D.1 ARCHITECTURE & TRAINING INFORMATION

The base ViT (Dosovitskiy et al., 2020) architecture has 6 transformer layers, embedding dimension (d) 512, and 8 attention heads per transformer layer. CIFAR-10 images of size 32×32 are processed as 4×4 patches. The Multilayer Perceptron (MLP) within each transformer block has hidden dimension 512, and the model uses learned positional embeddings.

The bottleneck architecture consists of a linear encoder followed by a linear decoder. The encoder projects pre-classification ViT activations into an m -dimensional latent space with $m \in \{2, 3, 5, 10\}$, and the decoder maps those compressed representations back to the $k = 10$ CIFAR-10 class logits. This gives a direct comparison across compression levels while holding the base model fixed.

The base ViT is trained on CIFAR-10 for 250 epochs with learning rate 0.001, Adam with default PyTorch parameters, batch size 512, cosine annealing, CE loss, and dropout. We train five random seeds to measure variability. After base training, the ViT weights are frozen and only the bottleneck is trained for 30 epochs with learning rate 0.001. Preprocessing uses RandomCrop with 4-pixel padding, resizing to the target size (32×32 by default), and RandomHorizontalFlip.

D.1.1 NORMALISED ROBUST ACCURACY ACROSS PERTURBATION MAGNITUDES

Fig. 4 (right) reports normalised robust accuracy for one ϵ value; this subsection expands the comparison across attack strengths. We report both ℓ_2 -norm (Tab. 7) and ℓ_∞ -norm (Tab. 6) PGD attacks, making clear how bottleneck dimension and perturbation magnitude jointly affect robustness.

For these evaluations, AExs are generated with PGD for 100 iterations using step size (α) 0.01. Robust accuracy is evaluated on 500 samples per configuration, and each table reports the mean normalised robust accuracy with standard deviation across five random seeds.

Table 6: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for CIFAR-10 ViT bottleneck models under ℓ_∞ -norm PGD attacks. Rows compare perturbation magnitudes (ϵ) and columns compare bottleneck dimensions (m); robust accuracy is normalised by each bottlenecked model’s clean accuracy. Across attack strengths, smaller m generally corresponds to lower robustness.

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.001	96.0% $\pm$ 0.4%	97.4% $\pm$ 0.7%	97.9% $\pm$ 0.4%	98.1% $\pm$ 0.3%
0.01	61.7% $\pm$ 3.6%	69.6% $\pm$ 3.5%	77.0% $\pm$ 0.9%	81.8% $\pm$ 3.2%
0.05	4.9% $\pm$ 1.5%	6.7% $\pm$ 1.6%	9.3% $\pm$ 0.6%	10.5% $\pm$ 0.8%
0.1	0.1% $\pm$ 0.2%	0.2% $\pm$ 0.3%	0.2% $\pm$ 0.4%	0.2% $\pm$ 0.4%
0.5	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%	0.0% $\pm$ 0.0%

D.1.2 ATTACK TRANSFERABILITY ACROSS PERTURBATION MAGNITUDES

We also report attack transferability across perturbation magnitudes (ϵ) and bottleneck dimensions (m), extending the transfer comparison in Sec. 4. Tab. 9 gives the ℓ_2 -norm transfer results, while Tab. 8 gives the ℓ_∞ -norm results, using the same source-seed to target-seed comparison structure.

Table 7: Mean normalised robust accuracy ($\pm$ standard deviation across 5 seeds) for CIFAR-10 ViT bottleneck models under ℓ_2 -norm PGD attacks. Rows vary perturbation magnitude (ϵ) and columns vary bottleneck dimension (m); robust accuracy is normalised by each model’s clean accuracy. The table preserves the main-text pattern: robustness decreases as attacks strengthen and as m becomes smaller.

ϵ	Bottleneck Dimension (m)			
	2	3	5	10
0.1	90.4% $\pm$ 1.7%	91.8% $\pm$ 0.7%	94.6% $\pm$ 1.2%	95.0% $\pm$ 0.7%
0.5	58.5% $\pm$ 5.0%	60.8% $\pm$ 5.0%	69.2% $\pm$ 1.8%	72.6% $\pm$ 2.8%
1.0	41.7% $\pm$ 4.8%	44.0% $\pm$ 3.0%	50.4% $\pm$ 0.6%	54.9% $\pm$ 1.7%
2.0	34.0% $\pm$ 4.6%	36.2% $\pm$ 3.3%	41.5% $\pm$ 1.3%	47.4% $\pm$ 2.4%
5.0	30.2% $\pm$ 4.1%	32.9% $\pm$ 3.5%	37.3% $\pm$ 1.9%	42.7% $\pm$ 1.0%

D.2 CORRELATIONS IN CLASS FEATURES

In Sec. 3, input correlations drive the superposition arrangements. In the CIFAR-10 bottleneck, the relevant correlations are instead correlations in the model’s class representations at the pre-classification layer. At this stage, the structure is closely related to the confusion matrix: classes that the model finds similar, and therefore tends to confuse, cluster together in the compressed representation. To test this interpretation, we repeat the experiment by fine-tuning a base ViT from `timm/vit_base_patch16_384`, pre-trained on ImageNet-21k (14 million images, 21,843 classes) and ImageNet (1 million images, 1,000 classes). After CIFAR-10 fine-tuning and the same bottleneck procedure as Sec. 4, the resulting geometry has random ordering across initialisations and approximately equal spacing between features (*i.e.* neuron collapse Kothapalli (2023)). In this high-performance regime, accuracy is near 100%, the confusion matrix is close to the identity, and the superposed geometry loses the structured clustering seen in lower-performing CIFAR-10-only models. By contrast, models trained solely on CIFAR-10 converge to similar geometries because they share the same difficulty structure: the same class pairs are hard to distinguish, producing consistent superposition patterns.

D.2.1 RESNET-18

To check architectural generalisation, we repeat the same bottleneck experiments with a ResNet-18 (He et al., 2016) base model instead of a ViT. The ResNet-18 reaches 92% clean accuracy, compared with 89% for the ViT in this comparison. Robust accuracy falls slightly faster for ResNet-18 at the same ϵ values, but the direction of the evidence is unchanged: normalised robust accuracy declines and transferability increases as bottleneck pressure rises.

E MODULAR ADDITION

Sec. 5 uses modular addition to test adversarial vulnerability when the relevant features are orthogonal rather than superposed. This section gives the experimental details, baseline comparisons, and certified-training results that support the algorithmic-brittleness interpretation.

E.1 LEARNED ALGORITHMIC REPRESENTATION IN MODULAR ADDITION

As detailed in Nanda et al. (2023) and Prieto et al. (2025b), the network solves modular addition by implementing a trigonometric algorithm. The encoder projects inputs onto non-basis-aligned linear representations encoding $\sin(\omega_k a)$ and $\cos(\omega_k a)$ at key frequencies $\omega_k = 2\pi k/P$. These features are orthogonal, *i.e.* not in superposition, and their sinusoidal values can be recovered with linear probes. The downstream MLP then applies sum-angle identities to compute $\cos((a + b)\omega_k)$ for several k , whose contributions constructively interfere at the logits to select the correct class. We generate AExs by relaxing the one-hot constraint and perturbing input $\mathbf{x}_j \in \mathbb{R}^P$ (encoding integer j) to $\mathbf{x}' = \mathbf{x} + \delta$ using ℓ_2 - and ℓ_∞ -norm PGD. For a controlled comparison, only a is attacked and b is held fixed.

Table 8: Attack transferability (%) for ℓ_∞ -norm PGD attacks on CIFAR-10 ViT bottleneck models. Each row fixes attack budget (ϵ) and bottleneck dimension (m); slash-separated entries show transfer from the source seed to the three listed target seeds. The ‘Mean $\pm$ Std’ column averages all 12 source–target seed pairings for each (ϵ, m) configuration, making the transfer comparison across compression levels explicit.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.001	2	77.8/66.7/44.4	25.0/62.5/25.0	72.7/72.7/72.7	58.3/75.0/58.3	59.3 $\pm$ 17.7
	3	57.1/28.6/28.6	14.3/28.6/28.6	63.6/72.7/72.7	42.9/42.9/42.9	43.6 $\pm$ 18.4
	5	70.0/50.0/50.0	57.1/71.4/57.1	50.0/25.0/25.0	16.7/50.0/33.3	46.3 $\pm$ 16.9
	10	55.6/55.6/66.7	12.5/37.5/25.0	16.7/33.3/16.7	28.6/71.4/28.6	37.3 $\pm$ 19.3
0.01	2	60.3/52.6/48.7	63.8/42.0/60.9	53.8/40.9/52.7	51.2/55.8/46.5	52.4 $\pm$ 6.9
	3	42.9/46.4/50.0	46.4/42.9/49.1	44.2/38.9/44.2	42.9/45.1/41.8	44.6 $\pm$ 3.0
	5	43.3/41.1/43.3	39.2/30.4/41.8	46.0/40.2/47.1	41.2/35.3/32.9	40.2 $\pm$ 4.8
	10	42.5/46.0/44.8	38.0/39.4/46.5	43.5/47.8/47.8	32.6/40.0/35.8	42.1 $\pm$ 4.7
0.05	2	35.0/36.4/37.8	38.8/36.0/46.3	37.4/31.3/41.7	42.0/42.0/40.7	38.8 $\pm$ 3.8
	3	29.9/29.9/37.1	30.4/30.1/35.6	30.2/30.2/35.1	30.1/31.9/34.4	32.1 $\pm$ 2.6
	5	26.8/25.3/29.2	25.2/21.5/26.4	28.0/28.0/26.5	26.0/26.6/21.9	25.9 $\pm$ 2.2
	10	27.1/24.9/30.4	21.6/23.0/28.4	25.3/28.1/30.6	21.0/24.9/20.1	25.4 $\pm$ 3.4
0.1	2	34.2/35.6/38.2	36.7/34.9/44.5	37.9/30.0/40.8	41.2/42.0/40.3	38.0 $\pm$ 3.8
	3	29.6/28.9/35.2	28.6/29.2/34.8	29.1/28.2/34.0	28.8/31.1/32.8	30.9 $\pm$ 2.5
	5	25.4/23.3/27.0	23.8/19.6/24.6	25.9/26.7/25.4	24.6/24.3/20.7	24.3 $\pm$ 2.1
	10	24.8/22.6/29.7	20.1/21.6/26.1	22.5/25.5/28.0	19.5/23.3/18.7	23.5 $\pm$ 3.3

Following Nanda et al. (2023), the learned W_e encodes sine and cosine values for key integer frequencies. These features are linearly decodable, and scaling a corresponding feature vector changes the encoded sinusoidal value rather than merely changing activation intensity. Integer embeddings W_e^j project onto a Fourier-like basis, producing a circular or periodic embedding geometry. Unlike the correlation-driven arrangements in the toy model, this circularity is imposed by the sine/cosine encoding of the non-correlated input integers.

Our model uses approximately seven key frequencies (ω_k). Each embedding W_e^j for an input integer x_j primarily encodes fourteen trigonometric features: $\cos(\omega_i x_j)$ and $\sin(\omega_i x_j)$ for $i \in \{1, 2, \dots, 7\}$. The concatenated 200-dim. representation therefore supplies ≈ 28 key trigonometric features to the MLP: fourteen for a and fourteen for b . The MLP combines them via sum-angle identities (e.g., $\cos(A + B) = \cos A \cos B - \sin A \sin B$) to compute $(x_1 + x_2) \pmod{P}$. The algorithmic brittleness analysed in Section 5 is the result of perturbing these specific learned trigonometric quantities.

E.2 EXPERIMENT DETAILS FOR MODULAR ADDITION TASK

The network is trained for addition modulo 113. Inputs are integer pairs (a, b) with $a, b \in \{0, \dots, 112\}$. Each integer is mapped to a 100-dimensional vector using a shared embedding $W_E \in \mathbb{R}^{113 \times 100}$, and W_E^a and W_E^b are concatenated into a 200-dim. feature vector. This vector is passed to a 3-hidden-layer MLP with 200 ReLU neurons per layer, and the output layer gives 113 classification logits. The model is trained end-to-end with cross-entropy loss and AdamW (weight decay 4), reaching 100% test accuracy. We generate AExs with PGD under ℓ_2 and ℓ_∞ norms across varying ϵ , perturbing W_E^a while holding b fixed to cause misclassification of $(a + b) \pmod{113}$.

Table 9: Attack transferability (%) for ℓ_2 -norm PGD attacks on CIFAR-10 ViT bottleneck models. The source-seed to target-seed layout matches Tab. 8, allowing direct comparison of transfer rates across ϵ and m under the ℓ_2 norm.

ϵ	m	Seed 10 vs. Seeds 20/30/40	Seed 20 vs. Seeds 10/30/40	Seed 30 vs. Seeds 10/20/40	Seed 40 vs. Seeds 10/20/30	Mean $\pm$ Std
0.1	2	70.6/64.7/41.2	60.0/50.0/60.0	64.3/60.7/57.1	48.1/63.0/51.9	57.6 ± 8.0
	3	39.1/39.1/47.8	50.0/60.7/50.0	48.1/55.6/51.9	48.0/28.0/40.0	46.5 ± 8.4
	5	43.5/34.8/43.5	31.6/31.6/42.1	42.9/35.7/21.4	30.4/39.1/34.8	35.9 ± 6.4
	10	45.5/59.1/45.5	19.0/28.6/28.6	37.5/43.8/43.8	23.8/52.4/38.1	38.8 ± 11.4
0.5	2	59.6/52.1/48.9	60.8/35.4/60.8	54.1/39.4/51.4	43.8/49.5/46.7	50.2 ± 7.7
	3	38.7/40.6/48.1	36.1/36.8/41.7	37.4/36.5/43.5	34.4/34.4/36.7	38.7 ± 3.9
	5	38.4/32.0/37.6	29.4/25.7/32.1	39.1/33.9/38.3	31.8/29.9/32.7	33.4 ± 4.0
	10	38.3/37.4/37.4	26.5/28.4/31.4	32.4/35.3/35.3	23.4/28.2/25.0	31.6 ± 5.0
1.0	2	50.8/43.7/44.4	51.7/36.4/50.0	46.1/32.9/44.7	42.7/46.7/45.3	44.6 ± 5.3
	3	35.7/36.9/41.7	31.8/36.4/42.6	33.7/32.0/37.3	34.9/32.0/35.5	35.9 ± 3.3
	5	32.1/28.9/33.2	30.9/23.0/30.9	35.9/31.5/33.7	27.7/28.2/27.1	30.3 ± 3.3
	10	32.3/32.3/38.5	23.7/27.7/29.9	29.1/32.0/34.3	22.2/25.6/21.1	29.1 ± 5.0
2.0	2	47.0/41.6/44.3	48.2/36.2/48.2	43.9/33.5/42.1	43.4/43.4/42.8	42.9 ± 4.2
	3	31.9/37.7/40.3	31.1/35.6/40.6	32.6/31.1/36.8	35.4/33.3/34.3	35.1 ± 3.2
	5	29.9/25.4/32.1	26.3/22.1/26.7	32.3/31.8/29.5	27.9/27.9/23.6	28.0 ± 3.2
	10	30.8/30.4/35.7	22.5/24.9/28.6	24.8/29.7/32.7	20.4/26.1/19.0	27.1 ± 4.9
5.0	2	44.7/46.7/40.8	45.3/37.3/48.7	42.7/33.9/41.5	42.3/42.3/40.6	42.2 ± 3.8
	3	32.3/35.8/40.8	31.3/32.6/39.9	31.0/30.0/33.8	35.0/33.0/33.0	34.1 ± 3.2
	5	30.6/24.1/28.6	28.3/20.9/24.8	31.7/31.3/27.4	28.1/29.0/23.2	27.3 ± 3.3
	10	29.8/27.4/32.7	19.7/24.1/28.9	24.9/29.9/31.2	18.8/25.8/19.2	26.0 ± 4.6

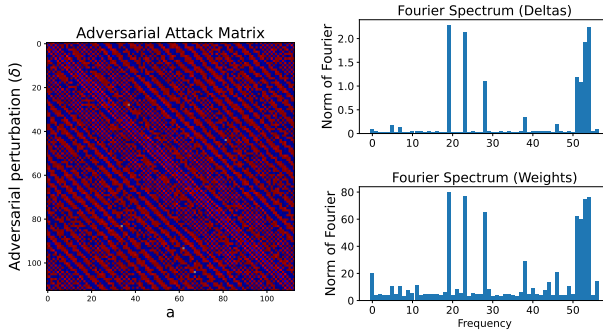


Figure 6: Frequency structure of successful modular-addition attacks. Left: each column is the mean perturbation δ added to a for successful attacks on $a + b \pmod p$. Right: Fourier-component norms of δ align with the key frequencies used by $W_{MLP_{out}}$.

Table 10: Minimum ℓ_2 -norm budget required to successfully attack $0 + 0 \pmod p$ for PGD, the informed frequency attack, and naive baselines.

Attack Method	Success ϵ
PGD	0.17
Informed Attack	0.22
Single a	0.97
ℓ_∞ Noise	2.04
Uniform δ	6.20

E.3 ALGORITHMIC BRITTLINESS

Frequency-aligned perturbations cause misclassification by changing the sinusoidal values read from the encoder features. For intuition, consider the hypothesised neuron-aligned case: $\mathbf{W}_e$ would contain one row for the sine or cosine at each ω_k , and the element at column $\mathbf{w}_e^a$ would contain $\sin(\omega_k a)$ or

$\cos(\omega_k a)$. For input a , the one-hot $\mathbf{x}_a$ selects $\mathbf{w}_e^a$, giving $\mathbf{h}_a = \mathbf{w}_e^a$. The perturbation δ changes $\mathbf{x}_a$ into a dense vector $\mathbf{x}'_a$, so $\mathbf{h}'_a = \mathbf{W}_e \mathbf{x}'_a = (1 + \epsilon_a) \mathbf{h}_a + \sum_{k \neq a} \epsilon_k \mathbf{w}_e^k$. The added $k \neq a$ sinusoidal terms corrupt the values used by the MLP and lead to misclassification. Because the algorithm relies on constructive interference of each ω_k at the logits, disrupting one frequency is insufficient; effective attacks must alter multiple frequency components together.

E.4 ANALYTIC CONSTRUCTION OF ATTACKS AND BASELINE COMPARISONS

The main paper shows that understanding the learned trigonometric algorithm makes it possible to construct attacks analytically. Sec. 5 benchmarks this ‘informed attack’ against PGD and naive baselines using the minimum successful ℓ_2 budget. This section defines the compared attack methods:

- **PGD Attack:** Standard PGD, used as the strongest optimisation baseline.
- **Informed Attack:** The analytically constructed perturbation formed from combinations of cosines at the learned frequencies.
- **Single a :** The minimum perturbation to a single input coordinate that causes misclassification.
- **ℓ_∞ Noise:** Random noise δ_{noise} added to input a , with each component sampled independently and uniformly from $[-\epsilon, \epsilon]$; the reported value is the ϵ that induces misclassification.
- **Uniform δ :** A perturbation vector with all components equal, i.e., $\delta_j = c \cdot \mathbf{1}$; the reported value is the magnitude of c required for misclassification.

E.5 DETAILS ON CERTIFIED ROBUST TRAINING

To evaluate whether defences change the identified mechanism or mainly shift attack budgets, we perform certified robust training against ℓ_∞ -norm perturbations. We use RSIP-IBP (Zhang et al., 2020; Henriksen & Lomuscio, 2023), an abstraction-based method (Wong & Kolter, 2018; Raghunathan et al., 2018; Palma et al., 2022) combining Interval Bound Propagation (IBP) (Katz et al., 2017; Ehlers, 2017) with Reversed Symbolic Interval Propagation (RSIP) to compute reachable output sets and certify robustness within a specified perturbation budget. The training loss includes both standard and RSIP components. Certified training runs for 1000 epochs, with the first 200 epochs used as warm-up; the learning rate is 1×10^{-3} and weight decay is 1×10^{-4} . We use the same train/test splits as standard training. Tab. 11 compares training budgets (ϵ_{train}) and evaluation budgets (ϵ_{attack}).

Table 11: Verified robust accuracy (%) for modular addition under ℓ_∞ -norm perturbations. The table compares the standard pre-robustification model with RSIP-IBP-trained models at different training budgets (ϵ_{train}), evaluated across attack budgets (ϵ_{attack}). Certified training improves robustness at covered budgets while leaving the same frequency-based sensitivity available at larger budgets.

Attack Epsilon (ϵ_{attack})	Train Epsilon (ϵ_{train})	Standard Acc. (Pre-Rob., %)	Robust Acc. (Pre-Rob., %)	Standard Acc. (Robust, %)	Robust Acc. (Robust, %)
5.0×10^{-4}	10^{-4}	100.0	100.0	100.0	100.0
	10^{-3}				100.0
1.0×10^{-3}	10^{-4}	100.0	76.0	100.0	93.8
	10^{-3}				100.0
2.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				95.7
5.0×10^{-3}	10^{-4}	100.0	0.0	100.0	0.0
	10^{-3}				85.3

As discussed in Sec. 5, subsequent PGD attacks on the robustly trained models still exploit the same key frequency components of the learned algorithm. The main effect of certified training is therefore quantitative rather than mechanistic: it increases the perturbation magnitude required for misclassification at certified budgets, but it does not remove the attack vector’s alignment with the model’s algorithmic sensitivities.

E.6 ON A POTENTIAL RECONCILIATION WITH SUPERPOSITION VULNERABILITY

Although this vulnerability arises without feature superposition, it resembles the superposition-driven case in one quantitative respect: both depend on sparsity assumptions in the input representation. This motivates our *reconciliation conjecture*: algorithmic vulnerabilities can persist in networks with orthogonal downstream representations when upstream computations violate sparsity. If earlier layers produce noisy or non-sparse features, potentially through adversarial interference, those deviations can enter otherwise well-isolated downstream circuits as effectively pre-perturbed inputs. Such circuits may operate correctly on clean inputs but fail once the upstream representation has shifted. This gives a plausible route by which algorithm-disruption mechanisms, including the frequency attacks studied here, could be activated even without local superposition. The interaction between upstream superposition, downstream algorithmic brittleness, and layer-by-layer vulnerability mechanisms remains underexplored, and a detailed empirical and theoretical investigation is a promising future direction.

F INSPECTING ADVERSARIAL PERTURBATIONS VIA SAE FEATURES

The main paper focuses on settings where the compared features are explicit class features in the toy model or class-level bottleneck representations in the ViT experiments. In large practical networks, however, interference can occur among unlabelled latent features across many layers. As discussed in the future-work section, SAEs provide an unsupervised route to extracting more linear features from NNs (Bricken et al., 2023). This appendix gives a preliminary, explicitly exploratory comparison of SAE-extracted features for clean, adversarial, and noisy inputs in a large-scale ViT. The goal is not to introduce a definitive new result, but to assess whether the same evidence style – comparing adversarial changes against feature-level measurements – can scale beyond labelled class features. We first describe the setup, then validate SAE behaviour on adversarial inputs, and finally present layer-wise feature-difference visualisations.

F.1 EXPERIMENT SETUP

We analyse SAE-extracted features from the vision encoder of LAION/CLIP-ViT-B-32-DataComp.XL-s13B-b90K, a 12-layer ViT with $\approx 70\%$ zero-shot accuracy on ImageNet-1k Deng et al. (2009). We use open-source Prisma SAEs that map the 768-dimensional activation space to 49,152 dictionary features ($64\times$ expansion) (Joseph et al., 2025). We compare SAEs trained on mean patch-token representations and CLS-token representations across all layers. To separate large activation changes from broader feature-set changes, we analyse a high threshold (0.1) and a low threshold (0.001). The dataset for these comparisons contains 10,000 original/successfully attacked image pairs generated with ℓ_∞ -norm attacks at $\epsilon = 0.02$.

F.2 SAE VALIDITY FOR ADVERSARIAL INPUTS

Because AExs are out-of-distribution relative to clean training data, we first validate whether standard SAEs remain useful probes under adversarial inputs. We insert SAEs one layer at a time during inference and compare model accuracy for clean, adversarial, and noisy inputs. This check asks whether the SAEs still process classification-relevant features under attack before we interpret differences in feature activations.

F.3 LAYER-WISE SAE FEATURE DIFFERENCES

This subsection compares SAE feature activations across ViT layers for original, ℓ_∞ adversarial, and random noisy inputs. The visualisations are exploratory and are intended to identify which feature-level metrics may be useful for future adversarial analyses. We examine:

- Activation count: Histograms comparing how many SAE features activate above threshold for original, adversarial, and noisy images, testing whether attacks alter sparsity.
- Feature overlap: Counts of common activated features for original vs. adversarial and original vs. noisy conditions, testing whether attacks activate a distinct feature set.
- Jaccard Similarity: A normalised overlap metric for the activated feature sets.

Table 12: ImageNet-1k top-1 accuracy when inserting a pre-trained SAE, one layer at a time, during inference for original, ℓ_∞ adversarial ($\epsilon = 0.02$), and random noisy inputs. ‘Original input + SAE’ means the clean ViT activation at that layer is reconstructed by the SAE before being passed onward. The small improvement for ‘Adv. input + SAE’ relative to ‘Adv. input’ suggests partial filtering, but the model still largely misclassifies, so adversarially relevant changes remain present after reconstruction.

Metric	Layer 2	Layer 5	Layer 8	Layer 11
Accuracy				
Original input	1.00	1.00	1.00	1.00
Original input + SAE	0.99	0.99	0.99	0.86
Adv. input	0.00	0.00	0.00	0.00
Adv. input + SAE	0.02	0.01	0.01	0.11
Noisy input	0.98	0.98	0.98	0.98
Noisy input + SAE	0.98	0.98	0.98	0.85
SAE Improvement				
Original input	-0.01	-0.01	-0.01	-0.14
Adv. input	0.02	0.01	0.01	0.11
Noisy input	-0.00	-0.00	-0.00	-0.13

- Mean activation: Histograms of activation magnitudes for features above threshold, comparing feature intensity across conditions.
- Distinct Features: Counts of unique features activated across each image set and condition.

F.4 CLS TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection analyses CLS-token SAE features using the high activation threshold 0.1. Figures 7 through 10 compare activation counts, feature overlap, Jaccard similarity, and mean activations for original, adversarial, and random noisy inputs across ViT layers.

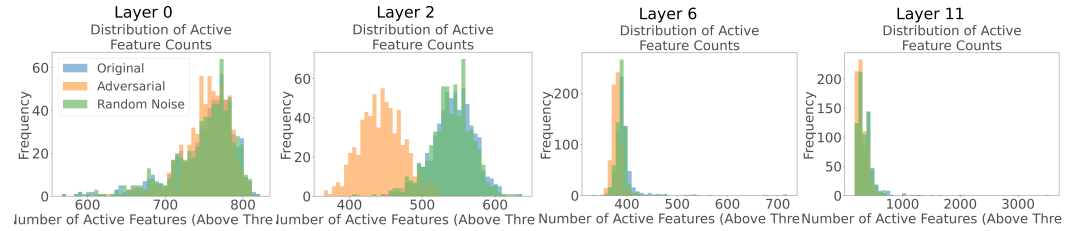


Figure 7: CLS-token SAE activation-count comparison at threshold 0.1 across original, adversarial, and noisy inputs.

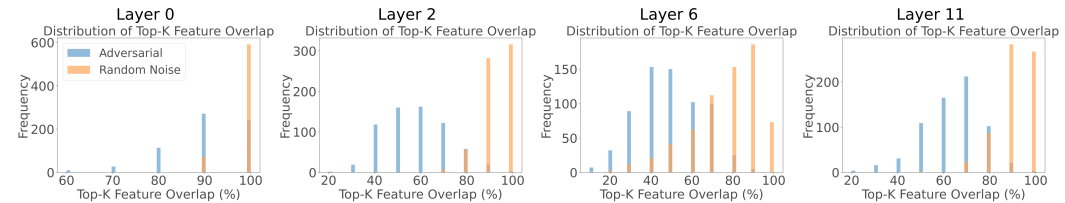


Figure 8: CLS-token SAE feature-overlap comparison at threshold 0.1 for original vs. attacked and original vs. noisy images.

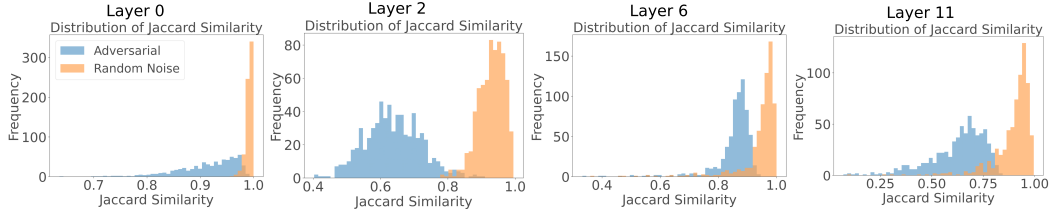


Figure 9: Jaccard-similarity comparison for CLS-token SAE features at threshold 0.1.

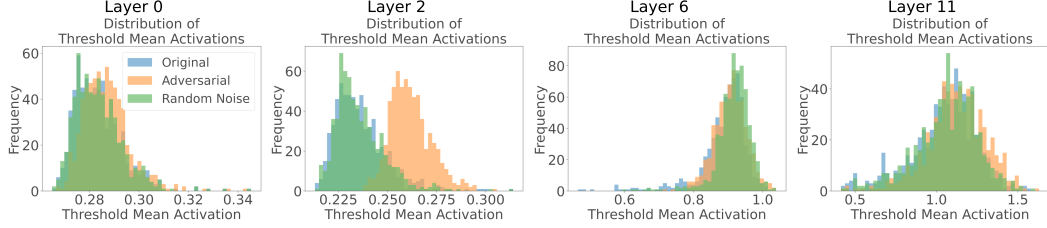


Figure 10: Mean-activation histogram for CLS-token SAE features at threshold 0.1.

F.5 CLS TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

We then lower the CLS-token activation threshold to 0.001 to include weaker but potentially informative features. The figures compare activation counts (Fig. 11), feature overlap (Fig. 12), Jaccard similarity (Fig. 13), and mean activations (Fig. 14) across layers for original, adversarial, and noisy inputs.

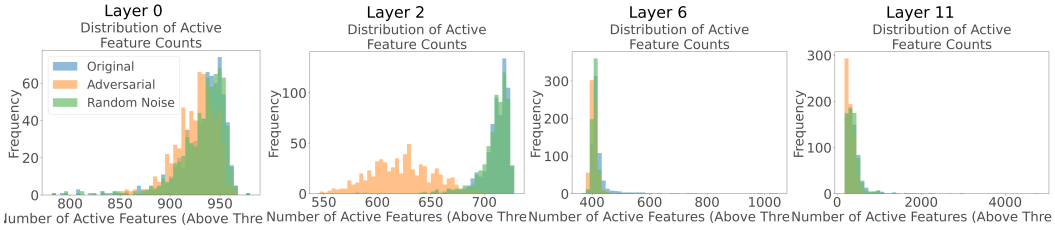


Figure 11: CLS-token SAE activation-count comparison at threshold 0.001, including noisy inputs.

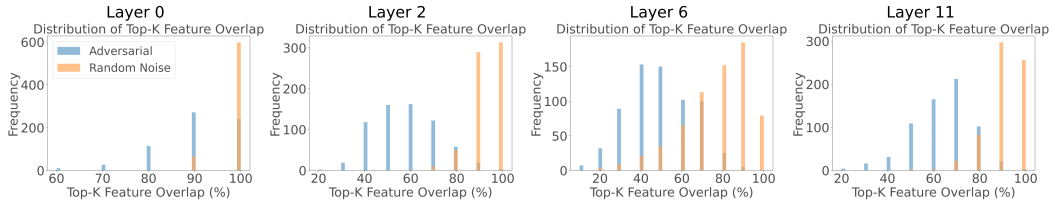


Figure 12: CLS-token SAE feature-overlap comparison at threshold 0.001 for original vs. attacked and original vs. noisy images.

F.6 PATCH TOKEN SAE FEATURES WITH HIGH ACTIVATION THRESHOLD (0.1)

This subsection shifts to SAE features derived from mean patch tokens, again using the high threshold 0.1. We compare activation counts (Fig. 15), distinct feature counts (Fig. 16), feature overlap (Fig. 17), Jaccard similarity (Fig. 18), and mean activations (Fig. 19) for original, adversarial, and noisy inputs.

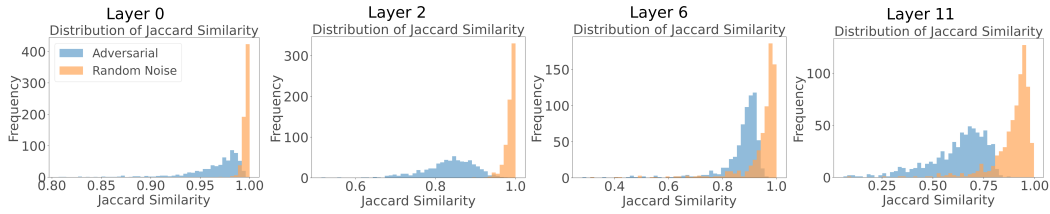


Figure 13: Jaccard-similarity comparison for CLS-token SAE features at threshold 0.001.

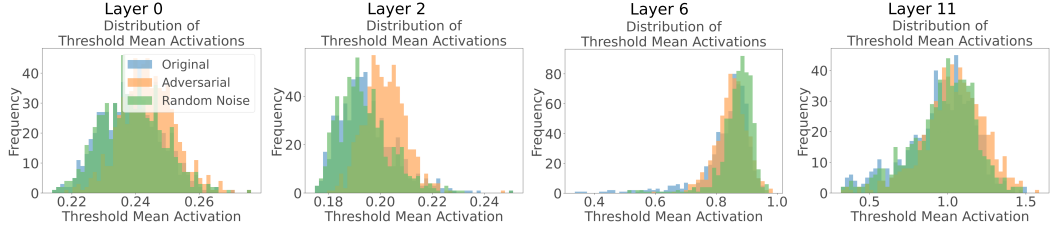


Figure 14: Mean-activation histogram for CLS-token SAE features at threshold 0.001, including noisy inputs.

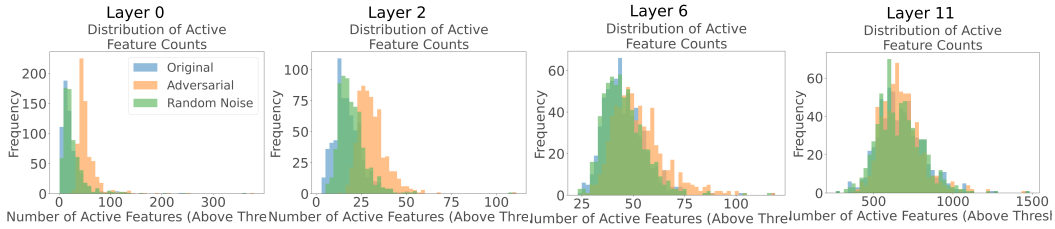


Figure 15: Patch-token SAE activation-count comparison at threshold 0.1, including noisy inputs.

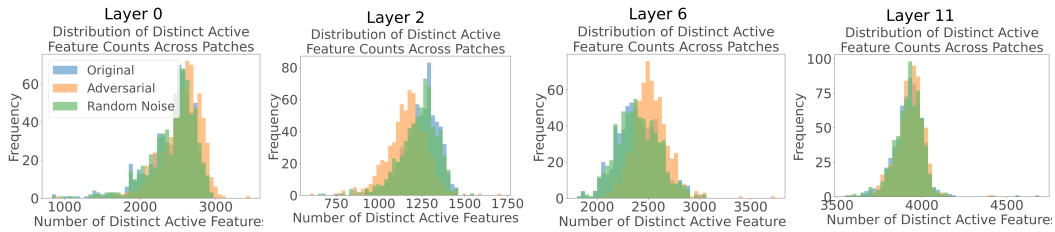


Figure 16: Distinct patch-token SAE feature counts at threshold 0.1, including noisy inputs.

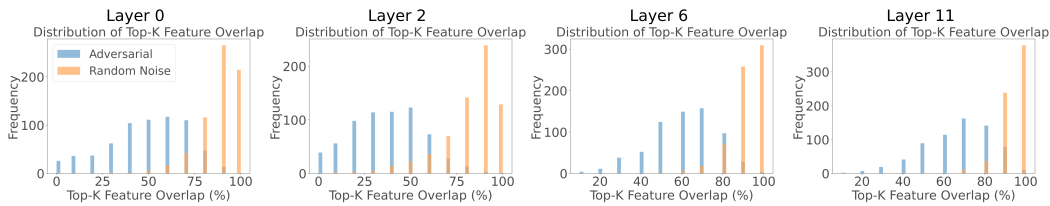


Figure 17: Patch-token SAE feature-overlap comparison at threshold 0.1 for original vs. attacked and original vs. noisy images.

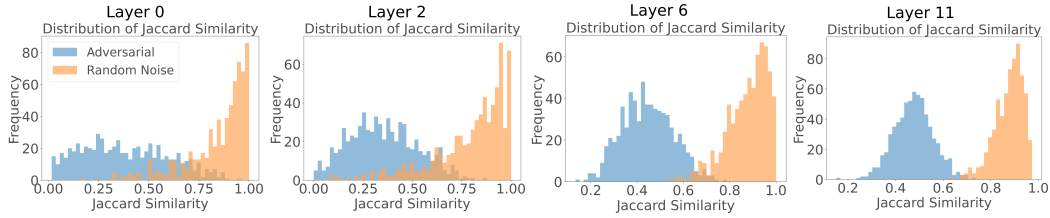


Figure 18: Jaccard-similarity comparison for patch-token SAE features at threshold 0.1.

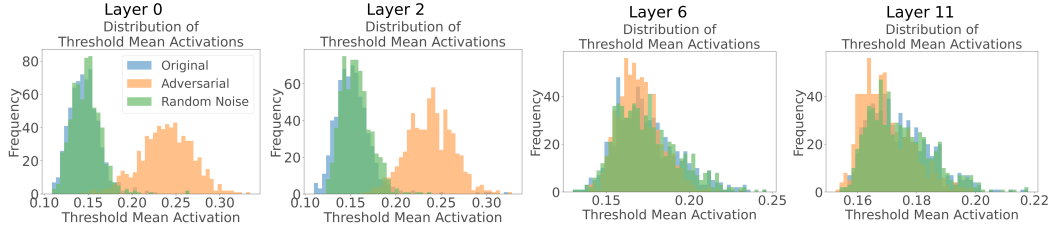


Figure 19: Mean-activation histogram for patch-token SAE features at threshold 0.1, including noisy inputs.

F.7 PATCH TOKEN SAE FEATURES WITH LOW ACTIVATION THRESHOLD (0.001)

Finally, we analyse mean patch-token SAE features at the low threshold 0.001. The figures compare activation counts (Fig. 20), distinct feature counts (Fig. 21), feature overlap (Fig. 22), Jaccard similarity (Fig. 23), and mean activations (Fig. 24) for original, adversarial, and noisy inputs.

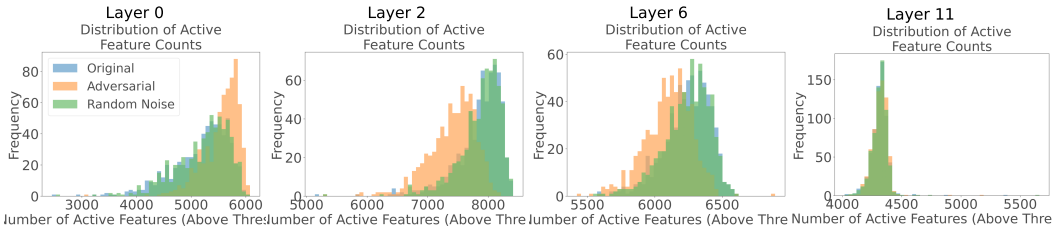


Figure 20: Patch-token SAE activation-count comparison at threshold 0.001, including noisy inputs.

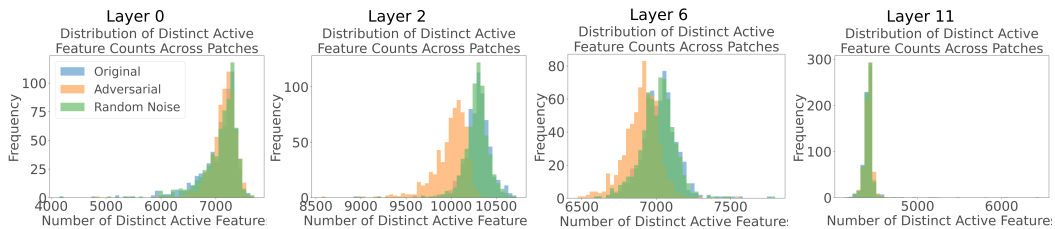


Figure 21: Distinct patch-token SAE feature counts at threshold 0.001, including noisy inputs.

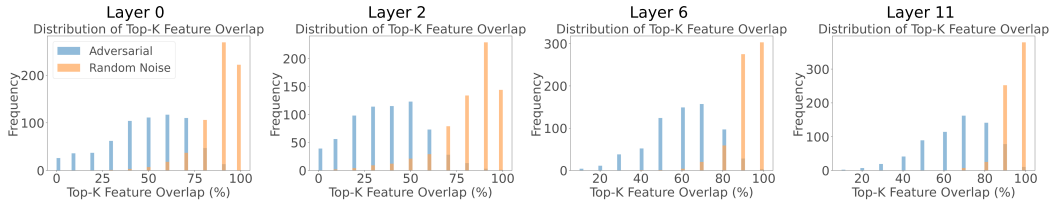


Figure 22: Patch-token SAE feature-overlap comparison at threshold 0.001 for original vs. attacked and original vs. noisy images.

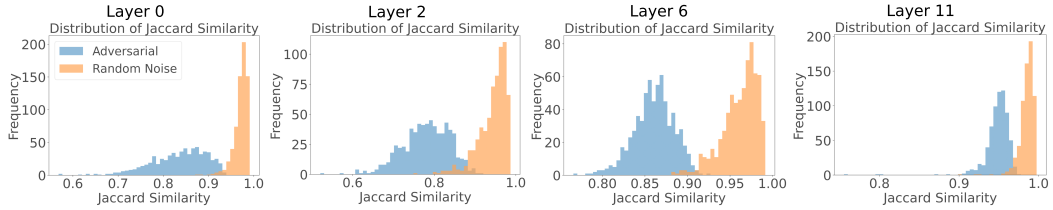


Figure 23: Jaccard-similarity comparison for patch-token SAE features at threshold 0.001.

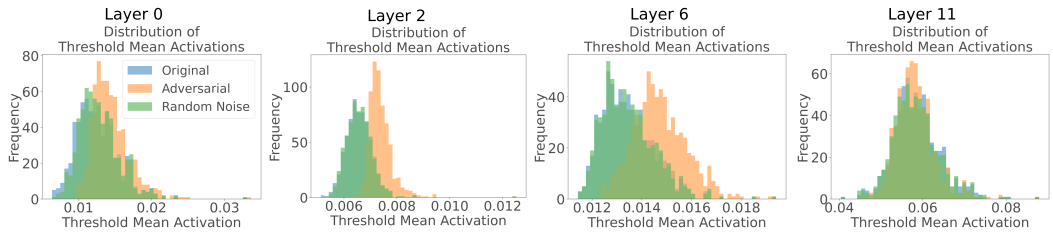


Figure 24: Mean-activation histogram for patch-token SAE features at threshold 0.001, including noisy inputs.

G ACRONYMS

AEx adversarial example

CE cross entropy

LRH linear representation hypothesis

MLP Multilayer Perceptron

MSE mean squared error

NN neural network

PGD projected gradient descent

SAE sparse autoencoder

ViT vision transformer